

# Exploration of three configurations of lines in surfaces

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60 ECTS study points

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## **Abstract**

This thesis studies configurations of lines contained in algebraic surfaces in projective three-space  $\mathbb{P}^3$ . Several new configurations are defined and studied, and the existence of inclusions of these configurations in surfaces of varying degree is investigated, using both theoretical and computational methods.

## **Sammendrag**

Denne oppgaven studerer konfigurasjoner av linjer som ligger på algebraiske flater i det prosjektive rommet  $\mathbb{P}^3$ . Flere nye konfigurasjoner blir definert og studert, og eksistensen av inklusjoner av disse konfigurasjonene i flater av ulik grad blir undersøkt, ved bruk av både teoretiske og numeriske resultater.

## **Declaration of AI assistance**

In this scientific work, generative artificial intelligence (AI) has been used. All data and personal information have been processed in accordance with the University of Oslo's regulations, and I, the author of the document, take full responsibility for its content, claims, and references. An overview of the use of generative AI is provided below.

This thesis is my own work. I have used AI tools (mostly Claude and ChatGPT) to support language and presentation, and specifically I have used it for:

- Improving grammar and style in sections of the text.
- Generating suggestions for rephrasing sentences or paragraphs for clarity.
- Assisting with formatting LaTeX code snippets.

I have not used AI tools to generate original scientific content, mathematical proofs, or research findings. All mathematical results, proofs, and analyses are my own work. No AI written sections have been included without proper review and modification by me. No AI tools have been used to generate references or citations, or to verify the accuracy of scientific content.

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# Preface

This thesis has in a way been found through “experimental” mathematics, that is that especially in the final 6 months, we have used Macaulay2 to find.



# Acknowledgements

chapter:ack

I would like to thank my supervisor Jørgen Vold Rennemo for all the help, support and guidance during this project. The supervising meetings have been a joy, and I have learned so much in the past year.

I would like to thank the entire 11th floor of the Mathematics Department. This is a special place filled with great people. I have learned so much from fellow students, the PhD candidates, and the professors, and the time spent here has been so much fun thanks to all of you. A special thanks to Aleksander Leraand, who went from being an inspiration as a senior master student, to being the biggest aid in my thesis apart from my supervisor. Your help has been invaluable.

I would also like to thank my family and friends for all the support and encouragement during my studies. I would especially like to thank my wonderful girlfriend Elin for her love, patience, and understanding throughout this journey. This thesis would not have been possible without her.





# Chapter 1

## Introduction

intro

In 1849, Arthur Cayley communicates to George Salmon that a general cubic surface contains a finite number of lines, and later Salmon responded it must be equal to 27. Later this same year, Salmon proves that in fact any non-singular surface contains exactly 27 lines. Dolgachev claims this can be considered the beginning of modern algebraic geometry ([Dol04]).

Fortsett.

### 1.1 Outline

There are two main aims of this thesis. One is proving the theorem that there are 27 lines on a cubic surface. The second aim is to study three line configurations on surfaces, which we define in this paper. These are the  $m$ -crosses,  $m$ -gons, and  $m$ -chains, and we provide explicit computational examples illustrating the phenomena discussed. We describe the theory of lines and higher dimensional subspaces of projective varieties to aid in our goals.

The paper is organized as follows:

- Chapter <sup>background</sup>1 collects some necessary theorems, and introduces intersection theory.
- Chapter <sup>grassmannians</sup>2 introduces the Grassmannian, its universal bundles, and the geometry of the Fano variety of lines.
- Chapter <sup>chern</sup>3 develops Chern class techniques and uses them to give enumerative proofs, culminating in several proofs that a smooth cubic surface contains exactly 27 lines.
- Chapter <sup>configs</sup>4 introduces  $m$ -crosses,  $m$ -gons and  $m$ -chains of lines on surfaces, studies existence and classification results, and relates these configurations to the general theory. We prove the novel result that a general 4-gon and 6-gon is only included in unique smooth surfaces of degree 4 and 6 respectively.
- The final section summarizes the results and lists open questions and possible directions for further work, as well as conjectures based on computational evidence.
- Appendix <sup>appendix</sup>A describes the computational methods and implementations used (computer algebra systems and scripts), and presents concrete examples and data.

Fjern punktliste.

## 1.2 Notation

We will use the following notation throughout the thesis:

- We work over the field of complex numbers  $\mathbb{C}$  unless otherwise stated.
- $K(X)$  denotes the function field of a variety  $X$ .
- $K(X)^\times$  denotes the multiplicative group of non-zero elements in  $K(X)$ .
- We denote  $\dim_{\mathbb{C}} H^i(-, -)$  by  $h^i(-, -)$ .

## 1.3 Important results

Start with some notation and conventions that will be used throughout the thesis.

We will now present some important results that will be used later in the thesis.

thm:bertini

**Theorem 1.3.1** (Bertini). *If  $\mathcal{D}$  is a linear system on a variety  $X$  in characteristic 0, then the general member of  $\mathcal{D}$  is smooth away from the base locus of  $\mathcal{D}$  and the singular locus of  $X$ .*

Add notation (this is from EH): A *linear system* on a scheme  $X$  is a pair  $\mathcal{D} = (\mathcal{L}, V)$ , where  $\mathcal{L}$  is a line bundle on  $X$  and  $V \subseteq H^0(X, \mathcal{L})$  a vector space of sections.

We refer to (!?) for a proof. <sup>[EH16]</sup>[EH16]

Find a good reference for this theorem.

thm:bezout

**Theorem 1.3.2** (Bézout). *Let  $X, Y \subseteq \mathbb{P}^n$  be two closed subvarieties having complementary dimension, i.e.  $\dim(X) + \dim(Y) = n$ . If  $X$  and  $Y$  intersect transversely, then they will intersect in exactly  $\deg(X) \cdot \deg(Y)$  points. Furthermore, if the intersection is finite, then the number of intersection points (counted with multiplicity) is at most  $\deg(X) \cdot \deg(Y)$ .*

We refer to (!?) for a proof. <sup>[EH16]</sup>[EH16] corollary 2.4.

Find a good reference for this theorem. Don't require transverse intersections.

If  $X$  and  $Y$  intersect in finitely many points, then the number of intersection points is at most  $\deg(X) \deg(Y)$ .

**Theorem 1.3.3** (<sup>[Vak25]</sup>[Vak25] 11.4.1). *Suppose  $\pi: X \rightarrow Y$  is a morphism of irreducible  $k$ -varieties, with  $\dim X = m$  and  $\dim Y = n$ . Then there exists a nonempty open subset  $U \subseteq Y$  such that for all  $q \in U$ , the fiber over  $q$  has pure dimension  $m - n$ , or is empty.*

irred\_fiber

**Lemma 1.3.4.** *Let  $\pi: X \rightarrow Y$  be a proper morphism to an irreducible variety, and all the fibers are nonempty and irreducible of the same dimension. Then  $X$  is irreducible.*

This can be found in <sup>[EO25]</sup>[EO25], exercise 14.2.21.

Sjekk at oppgaven er samme plass.

thm:fiber

**Theorem 1.3.5** (Dimension formula for fibers). *Let  $k$  be a field and let  $f: X \rightarrow Y$  be a dominant morphism of irreducible projective varieties over  $k$ . Let  $\eta$  denote the generic point of  $Y$  and write  $X_\eta = X \times_Y \operatorname{Spec} k(Y)$  for the generic fiber. Then*

$$\dim X = \dim Y + \dim X_\eta.$$

Write the proof, and/or find reference. Maybe use <sup>[Vak25]</sup>[Vak25] 11.4.2.



## Chapter 2

# Grassmannians and the space of lines in $\mathbb{P}^3$

assmannians

In this chapter we will introduce Grassmann varieties, or Grassmannians, and focus on the space of lines in  $\mathbb{P}^3$ , which is denoted  $\mathbb{G}(1, 3)$ . We will also describe the Grassmannians as projective varieties by introducing its affine cover, and a way to embed the Grassmannian into a projective space, giving it the structure of a variety (gjentakelse!). For a more detailed introduction to Grassmannians, we refer to Chapter 3 and 4 in [EH16], or [EO25, Section 24.6]. We closely follow the exposition in [EO25].

### 2.1 Definition and basic properties

A Grassmannian is a projective variety whose closed points correspond to linear subspaces of a certain dimension in a vector space. As a set,  $\text{Gr}(k, V)$  is the set of all  $k$ -dimensional linear subspaces of the  $n$ -dimensional vector space  $V$ . In our case,  $V$  will be a vector space over an algebraically closed field  $\mathbb{C}$ .

**Definition 2.1.1.** We define the *Grassmannian*  $\text{Gr}(k, V)$  of  $k$ -dimensional linear subspaces of  $V$  as the set

$$\text{Gr}(k, n)(\mathbb{C}) = \{[L] \mid L \subseteq \mathbb{C}^n \text{ is a linear space of dimension } k\}.$$

A  $k$ -dimensional linear subspace of an  $n$ -dimensional vector space  $V$  can also be thought of as a  $(k - 1)$ -dimensional linear subspace of  $\mathbb{P}V \cong \mathbb{P}^{n-1}$ . Due to this, we will often write  $\text{Gr}(k, V)$  as  $\mathbb{G}(k - 1, \mathbb{P}V)$ . When there is no need to specify the vector space  $V$ , only its dimension  $n$ , we will write  $\text{Gr}(k, n) = \mathbb{G}(k - 1, n - 1)$ . In this thesis we will most often write  $\mathbb{G}(k - 1, n - 1)$ , as it will be more natural when working with linear subspaces of projective space. For example,  $\mathbb{G}(1, 3)$  parametrizes lines in  $\mathbb{P}^3$ .

The Grassmannian is an important and very useful structure in algebraic geometry. Naturally, when asking questions regarding lines, planes, or other linear subspaces of projective spaces, many answers include the geometry of this variety of the linear subspaces themselves.

### 2.2 Covering by affine spaces

To describe the Grassmannian as a projective variety, we will describe an affine cover of the Grassmannian. Section 24.6 of [EO25] goes into detail about this construction, and

shows that the cover glues to a scheme. We will go over the overall idea, but exclude the finer details.

We represent a linear subspace  $L \subseteq \mathbb{C}^n$  as the kernel of a surjective linear map  $\mathbb{C}^n \rightarrow \mathbb{C}^{n-k}$ . This map can be represented by an  $(n-k) \times n$  matrix  $A$  of rank  $(n-k)$ . We have  $L = \text{Ker}(A) \subseteq \mathbb{C}^n$ , where

$$A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{(n-k),1} & \cdots & a_{(n-k),n} \end{pmatrix}.$$

Note that the matrix  $A$  is not unique for a given linear subspace  $L$ , since multiplying  $A$  on the left by an invertible  $(n-k) \times (n-k)$  matrix  $G \in \text{GL}_{n-k}(\mathbb{C})$  gives another matrix  $GA$  with the same kernel as  $A$ . For this reason, we consider the equivalence relation on the set of  $(n-k) \times n$  matrices of rank  $(n-k)$  given by  $A \sim A'$  if there exists an invertible  $(n-k) \times (n-k)$  matrix  $G$  such that  $A' = GA$ . The Grassmannian  $\text{Gr}(k, n)$  can then be identified with the set of equivalence classes of  $(n-k) \times n$  matrices of rank  $(n-k)$  under this equivalence relation. We notice that if  $L$  is represented by a matrix  $A$ , we can find a more canonical example by instead regarding  $A$  in its reduced echelon form. In this way, we represent  $L$  by a matrix  $A$  with some  $(n-k) \times (n-k)$  identity matrix as a submatrix. At the same time, a matrix  $A$  determines a subspace  $L \subseteq \mathbb{C}^n$ , and two matrices in reduced echelon form represent the same subspace if and only if they are equal. Note that a matrix in reduced echelon form with an  $(n-k) \times (n-k)$  identity matrix as a submatrix is parametrized by an affine space of dimension  $(n-k)k$ .

**Example 2.2.1.** The matrices

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

since they represent the same 2-dimensional subspace of  $\mathbb{C}^4$ , and is in the same class as the matrix their reduced echelon form

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

▲

Dette afsnittet kan komme før jeg snakker om echelon form.

For these reasons, we consider the set

$$(\mathbb{A}^{(n-k)n}(\mathbb{C}) - Z) / \text{GL}_{n-k}(\mathbb{C})$$

where  $\mathbb{A}^{(n-k)n}(\mathbb{C})$  parametrizes the set of all  $(n-k) \times n$  matrices with entries in  $\mathbb{C}$ , and  $Z$  is the closed subset of  $(n-k) \times n$  matrices of rank less than  $(n-k)$ . The  $\mathbb{C}$ -points of  $\text{Gr}(k, n)$  is bijective to the classes in this set. We now define an open cover of  $\text{Gr}(k, n)$ .

Let  $I \subseteq \{1, \dots, n\}$  be a subset of size  $(n-k)$ . For each  $I$  let  $U_I \subseteq \mathbb{A}^{(n-k)n}(\mathbb{C})$  denote the closed subscheme such that the columns given by  $I$  form a  $(n-k) \times (n-k)$  submatrix which is the identity matrix. Each  $U_I$  is isomorphic to the affine space  $\mathbb{A}^{(n-k)k}$ . The collection of all such  $U_I$  for all subsets  $I$  of size  $(n-k)$  gives an open cover of  $\text{Gr}(k, n)$ . Note that there are  $\binom{n}{n-k} = \binom{n}{k}$  such subsets, one for each  $I$ , which glues together to the Grassmannian the structure of a projective scheme.

**Example 2.2.2.** For  $n = 4, k = 2$ , there are 6 affine spaces  $\mathbb{A}^4$ . They are:

$$\begin{aligned} U_{12} &= \begin{pmatrix} 1 & 0 & a_{1,3} & a_{1,4} \\ 0 & 1 & a_{2,3} & a_{2,4} \end{pmatrix} & U_{13} &= \begin{pmatrix} 1 & a_{1,2} & 0 & a_{1,4} \\ 0 & a_{2,2} & 1 & a_{2,4} \end{pmatrix} \\ U_{14} &= \begin{pmatrix} 1 & a_{1,2} & a_{1,3} & 0 \\ 0 & a_{2,2} & a_{2,3} & 1 \end{pmatrix} & U_{23} &= \begin{pmatrix} a_{1,1} & 1 & 0 & a_{1,4} \\ a_{2,1} & 0 & 1 & a_{2,4} \end{pmatrix} \\ U_{24} &= \begin{pmatrix} a_{1,1} & 1 & a_{1,3} & 0 \\ a_{2,1} & 0 & a_{2,3} & 1 \end{pmatrix} & U_{34} &= \begin{pmatrix} a_{1,1} & a_{1,2} & 1 & 0 \\ a_{2,1} & a_{2,2} & 0 & 1 \end{pmatrix} \end{aligned}$$

▲

One can also show that this construction is possible over all rings by working over the base ring  $\mathbb{Z}$ . We will not go into details here, but refer to [EO25] section 24.6.

**Proposition 2.2.3.** *For a field  $\mathbb{C}$ , the Grassmannian  $\text{Gr}_{\mathbb{C}}(k, n)$  is a regular projective variety of dimension  $k(n - k)$ .*

*Proof.* By construction, the Grassmannian has an affine covering of affine spaces  $\mathbb{A}^{(n-k)k}$ , so it is regular of dimension  $k(n - k)$ .  $\square$

ex:G13

**Example 2.2.4.** The Grassmannian  $\mathbb{G}(1, 3)$  parametrizing lines in  $\mathbb{P}^3$  is a regular projective variety of dimension 4.  $\blacktriangle$

## 2.3 The Plücker embedding

sec:Plucker

Again we follow [EO25] in describing the Plücker embedding, and keep the exposition brief. We look at  $\mathbb{P}_{\mathbb{C}}^N$  for  $N = \binom{n}{k} - 1$ . Then there is a closed embedding

$$\rho: \text{Gr}(k, n) \rightarrow \mathbb{P}^{\binom{n}{k}-1},$$

called the *Plücker embedding*. Locally, consider the morphism

$$\rho_I: U_I \rightarrow \mathbb{P}^{\binom{n}{k}-1}$$

defined by setting for each ring  $R$ , and  $A \in U_I(R)$

$$\rho_I(A) = [\det(A_J)]_{J \subseteq \{1, \dots, n\}, |J|=k}.$$

That is,  $\rho_I$  sends a matrix to all of its  $(n - k) \times (n - k)$  minors. These morphisms  $\rho_I$  glue together to give the Plücker embedding  $\rho$ .

**Lemma 2.3.1.** *For each  $I$ , the morphism  $\rho_I$  is a locally closed embedding.*

The proof of this lemma can be found in [EO25], Lemma 24.24. This means that  $\text{Gr}(k, n)$  embeds as a closed subscheme of  $\mathbb{P}^{\binom{n}{k}-1}$ . The image of this embedding is described by the vanishing of certain homogeneous polynomials called the Plücker relations.

Legg til plucker relations?

## 2.4 Universal sub and quotient bundles

Later, we will need the notion of the universal subbundle over the Grassmannian. Here we follow the exposition in <sup>EH16</sup>[EH16] Section 3.2.3. Let  $V$  be an  $n$ -dimensional vector space over a field  $\mathbb{C}$ . Consider the Grassmannian  $\mathrm{Gr}(k, V)$  of  $k$ -dimensional linear subspaces of  $V$ . Let  $\mathcal{V} = \mathrm{Gr}(k, V) \times V$  be the trivial vector bundle over  $\mathrm{Gr}(k, V)$  with fiber  $V$ .

**Definition 2.4.1.** The *universal subbundle*  $\mathcal{S}$  over  $\mathrm{Gr}(k, V)$  is the vector bundle whose fiber over a point  $[L] \in \mathrm{Gr}(k, V)$  is the subspace  $L \subseteq V$ . That is,

$$\mathcal{S}_{[L]} = L \subseteq V = \mathcal{V}_{[L]}.$$

**Definition 2.4.2.** The quotient  $\mathcal{Q} = \mathcal{V}/\mathcal{S}$  is called the *universal quotient bundle* over  $\mathrm{Gr}(k, V)$ , and can be viewed as the vector bundle whose fiber over a point  $[L] \in \mathrm{Gr}(k, V)$  is the quotient space  $V/L$ .

**Proposition 2.4.3** (<sup>EH16</sup>[EH16], Proposition 3.3). *The subset  $\mathcal{S}$  of  $\mathcal{V}$  whose fiber over a point  $[L] \in \mathrm{Gr}(k, V)$  is the subspace  $L \subseteq V$  is a vector bundle over  $\mathrm{Gr}(k, V)$ .*

Later we will use the universal subbundle to prove that a smooth cubic surface contains exactly 27 lines.

## 2.5 Short on Schubert cycles (!?)

Bestem om vi skal inkludere. Er litt relevant for 27 linjer, og vi bruker teknisk sett Schubert varieteter senere.

- Consider adding examples from EH16 to illustrate the kinds of questions one can answer after this chapter.



## 2.6 Incidence correspondence and the universal Fano scheme

sec:fano

We aim to introduce the concept of incidence correspondences (or an incidence variety), which will be a tool we will use several times later in this thesis. We do this by introducing the universal Fano scheme, which is an example of an incidence variety. Generally, an incidence correspondence is a set of pairs  $(x, y)$  such that  $x$  lies in  $y$ , where  $x$  and  $y$  are elements of  $X$  and  $Y$  respectively, for some varieties  $X$  and  $Y$  parametrizing geometric objects. This can be visualized in the following diagram, with projection maps  $\pi_1$  and  $\pi_2$ :

$$\begin{array}{ccc} & \mathcal{I} = \{(x, y) \in X \times Y \mid x \text{ lies in } y\} & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ X & & Y \end{array} \quad (2.1)$$

Now we define an incidence correspondence of  $k$ -planes  $L \in \mathbb{G}(k, n)$  contained in degree  $d$  surfaces in  $\mathbb{P}^n$ .

**Definition 2.6.1.** ([EH16] Section 6.1). Let  $N = \binom{n+d}{d} - 1$ , so that  $\mathbb{P}^N$  parametrizes all degree  $d$  hypersurfaces in  $\mathbb{P}^n$ . The *universal Fano scheme* is defined as

$$\Phi(n, d, k) = \{(X, L) \in \mathbb{P}^N \times \mathbb{G}(k, n) \mid L \subseteq X\}$$

universal Fano

**Remark 2.6.2.** The *Fano scheme*  $F_k(X) \subseteq \mathbb{G}(k, n)$  of a variety  $X \subseteq \mathbb{P}^n$  parametrizes  $k$ -planes on  $X$ . The universal Fano scheme parametrizes  $k$ -planes for all varieties of a certain degree in  $\mathbb{P}^n$ . By taking the fiber over a point  $[X] \in \mathbb{P}^N$ , that being the class of the hypersurface  $X \subseteq \mathbb{P}^n$ , we get the Fano scheme  $F_k(X)$  for the specified variety. In this sense, taking the fiber over a “general”  $X$  results in a “general” Fano scheme  $F_k(X)$ . Studying the universal Fano scheme can then tell us what we can “expect” when examining a specific Fano scheme. It is therefore a very useful tool in telling us about a general member of the varieties.

**Lemma 2.6.3.**  $\Phi \subset \mathbb{P}^N \times \mathbb{G}(k, n)$  is a closed subset of  $\mathbb{P}^N \times \mathbb{G}(k, n)$ .

Decide if we want to prove, or refer to proof. Can refer to Harris 1995 chapter 6, or perhaps proof of 24.31 in EO

universalFano

**Proposition 2.6.4** ([EH16], Proposition 6.1). Let  $N = \binom{n+d}{d} - 1$ . The universal Fano scheme  $\Phi = \Phi(n, d, k) \subseteq \mathbb{P}^N \times \mathbb{G}(k, n)$  is a smooth irreducible variety of dimension

$$\dim \Phi(n, d, k) = N + (k+1)(n-k) - \binom{k+d}{k}.$$

We ask when we expect a general smooth surface  $X$  of degree  $d$  to contain lines, and remember that this question is answered by looking at the geometry of the Fano scheme  $F_k(X)$  for a general smooth surface  $X$ . More precisely one asks what dimension the Fano scheme has, therefore deciding what dimension the set of lines in our general surface is. As noted in Remark 2.6.2,  $F_k(X)$  can be obtained by taking the fiber over a point  $[X] \in \mathbb{P}^N$ . We therefore ask what dimension this fiber will have. From Theorem 1.3.5 we get that for a general point  $[X]$ ,  $\dim \pi^{-1}([X]) = \dim \Phi - \dim \mathbb{P}^N$  (unless  $\pi^{-1}([X])$  is empty)(Skriv om!?). We define this as  $\phi(n, d, k) := \dim \Phi - N$ . Now we can answer the question of when we expect lines to be contained in our surface by looking at  $\phi(n, d, k)$ :

**Corollary 2.6.5** (<sup>EH16</sup>[EH16], corollary 6.2). (a) *The dimension of any component of the family of  $k$ -planes on any hypersurface of degree  $d$  in  $\mathbb{P}^n$  is at least*

$$\phi(n, d, k) := (k + 1)(n - k) - \binom{k + d}{k}.$$

- (b) *If  $\phi(n, d, k) < 0$ , then the general hypersurface of degree  $d$  in  $\mathbb{P}^n$  contains no  $k$ -planes.*
- (c) *If  $\phi(n, d, k) \geq 0$ , and the general hypersurface of degree  $d$  contains any  $k$ -planes, then every hypersurface of degree  $d$  contains  $k$ -planes, and every component of the family of  $k$ -planes on a general hypersurface of degree  $d$  has dimension exactly  $\phi(n, d, k)$ .*

*Proof.*

Decide if we want to prove, or refer to proof.

□

To bring this back to a familiar context of lines on surfaces in  $\mathbb{P}^3$ , we set  $n = 3$  and  $k = 1$ . We then get that  $\phi(3, d, 1) = 3 - d$ . We use this in the following corollary:

**cor:Fano**

**Corollary 2.6.6.** *For a hypersurface  $X \subseteq \mathbb{P}^3$  of degree  $d$ , we expect to find no lines on  $X$  when  $d \geq 4$ , finitely many lines when  $d = 3$ , and positive-dimensional families of lines when  $d \leq 2$ .*

The proof of this can be found in <sup>E025</sup>[EO25], 22.31.

Decide if we want to prove, or refer to proof.

## Chapter 3

# Intersection theory and Chern classes

chern

### 3.1 Intersection Theory

An important tool in algebraic geometry is intersection theory. While we will only use it explicitly in a few cases in this thesis, it is a foundational concept which is useful for building intuition. We will follow the treatment of intersection theory of divisors as presented in the first chapter of [Bea96].

Let  $X$  be a smooth variety. Recall that the Picard group  $\text{Pic}(X)$  is the group of isomorphism classes of line bundles  $\mathcal{L}$  (i.e. of invertible sheaves) on  $X$ , with addition given by the tensor product,  $[\mathcal{L}] + [\mathcal{L}'] = [\mathcal{L} \otimes \mathcal{L}']$ . With every divisor  $D$  on  $X$ , we associate a line bundle  $\mathcal{O}_X(D)$ , defined locally as the set of rational functions  $f$  on  $X$  such that  $\text{Div}(f) + D \geq 0$ . If  $D$  is effective,  $\mathcal{O}_X(D)$  comes with a canonical section  $s \in H^0(X, \mathcal{O}_X(D))$ ,  $s \neq 0$  which is unique up to scalar multiplication, such that  $\text{Div}(s) = D$ . The map  $D \mapsto \mathcal{O}_X(D)$  identifies  $\text{Pic}(X)$  with the group of linear equivalence classes of divisors on  $X$ .

Let  $Y$  be another smooth variety, and consider a morphism  $f: X \rightarrow Y$ . Then we can define the pullback of an invertible sheaf with respect to  $f$  ([EO25, Section 16.5]) to get a homomorphism  $f^*: \text{Pic}(Y) \rightarrow \text{Pic}(X)$ . If  $f$  is a dominant morphism, we can also define the pullback  $f^*D$  of a Cartier divisor  $D$  on  $Y$  as a Cartier divisor on  $X$  such that  $f^*(\mathcal{O}_Y(D)) = \mathcal{O}_X(f^*D)$ . See [EO25, Section 20.7] for details.

Let  $X, Y$  be surfaces (See kommentar!?). If  $f$  is generically finite of degree  $d$ , then we define the direct image  $f_*C$  of an irreducible curve on  $X$  by setting

$$f_*C := \begin{cases} 0, & \text{if } f(C) \text{ is a point,} \\ r\Gamma, & \text{if } f(C) = \Gamma \text{ is a curve and the induced map } C \rightarrow \Gamma \text{ is finite of degree } r. \end{cases}$$

We define  $f_*D$  for all divisors  $D$  on  $X$  by linearity.

We now define intersection multiplicity.

**Definition 3.1.1.** Let  $C, C'$  be two distinct irreducible curves on a surface  $S$ ,  $x \in C \cap C'$ ,  $\mathcal{O}_x$  the local ring of  $S$  at  $x$ . If  $f$  and  $g$  are local equations of  $C$  and  $C'$  in  $\mathcal{O}_x$ , then the *intersection multiplicity* of  $C$  and  $C'$  at  $x$  is defined as

$$m_x(C \cap C') = \dim_{\mathbb{C}} \mathcal{O}_x / (f, g).$$

**Definition 3.1.2.** If  $C, C'$  are two distinct irreducible curves on a surface  $S$ , then the *intersection number* of  $C$  and  $C'$  is defined as

$$C.C' = \sum_{x \in C \cap C'} m_x(C \cap C').$$

**Theorem 3.1.3.** *Let  $C, C'$  be two distinct irreducible curves on a surface  $S$ . Then*

$$\mathcal{O}_S(C) \cdot \mathcal{O}_S(C') = C \cdot C'.$$

Definer (eller påstå) produkt av linjebunter først.

*Proof.* See [Bea96, Theorem I.4] □

In other words, we can compute the intersection number of two curves by computing the intersection number of their associated line bundles. If  $D, D'$  are two divisors on  $S$  we write  $D \cdot D'$  for  $\mathcal{O}_S(D) \cdot \mathcal{O}_S(D')$ . Then we have that we can calculate this intersection number by replacing  $D$  or  $D'$  by linearly equivalent divisors. This allows us to for example compute the self-intersection number of a divisor, denoted  $D \cdot D = D^2$ . Next we give some applications of this, which also serve as instructive examples of working with intersection theory.

**Example 3.1.4** (Example I.9 (a) in [Bea96]). Let  $S = \mathbb{P}^2$ . We know that  $\text{Pic}(S) \cong \mathbb{Z}$ . Let  $C, C'$  be curves of degree  $d$  and  $d'$  respectively. Since  $C \equiv dL$  and  $C' \equiv d'L'$  for distinct lines  $L, L'$ , we get Bezout's theorem:

$$C \cdot C' = dL \cdot d'L' = dd'(L \cdot L) = dd'.$$

Skriv hvorfor  $L \cdot L = 1$ ?

▲

**Example 3.1.5** (Proposition I.8 (a) in [Bea96]). Let  $C$  be a smooth curve, and let  $f: S \rightarrow C$  be a surjective morphism from a smooth and projective surface  $S$  to  $C$ , with  $F$  a fiber of  $f$ . Then  $F^2 = 0$ . ▲

*Proof.*  $F = f^*[x]$  for some point  $x \in C$ . Then there exists a divisor  $A$  on  $C$  linearly equivalent to  $x$  (all points are linearly equivalent on a curve), such that  $x \notin A$ . Then  $F \equiv f^*A$ . Since  $f^*A$  is a linear combination of fibers of  $f$  all distinct from  $F$ , we have  $F^2 = F \cdot f^*A = 0$ . □

Now we consider the case of a blow-up. Let  $S$  be a smooth surface, and let  $\pi: \hat{S} \rightarrow S$  be the blow-up of  $S$  at a point  $p \in S$ . Let  $E = \pi^{-1}(p)$  be the exceptional divisor, which we recall is isomorphic to  $\mathbb{P}^1$ . Also recall that for any curve  $C$  on  $S$ , we can consider the strict transform  $\hat{C}$  of  $C$  on  $\hat{S}$ , defined as the closure of  $\pi^{-1}(C \setminus \{p\})$  in  $\hat{S}$ . Then we have the following lemma on the pullback of divisors under the blow-up map.

lem:strict

**Lemma 3.1.6** (Lemma II.2 in [Bea96]). *Let  $C$  be an irreducible curve on  $S$  which passes through  $p$  with multiplicity  $m$ . Then  $\pi^*(C) = \hat{C} + mE$* 

**Proposition 3.1.7** (Proposition II.3 (b) in [Bea96]). *Let  $S$  be a surface,  $\pi: \hat{S} \rightarrow S$  blow up of a point  $p \in X$  and  $E \subseteq \hat{S}$  the exceptional curve. Let  $D, D'$  be two divisors on  $S$ . Then  $(\pi^*D) \cdot (\pi^*D') = D \cdot D'$ ,  $E \cdot (\pi^*D) = 0$ ,  $E^2 = -1$*

*Proof.* For the first two equalities, we can replace  $D$  and  $D'$  by linearly equivalent divisors not passing through  $p$ , so that their strict transforms do not intersect  $E$ , and their intersection numbers can be computed on  $S$ . For the last equality, choose a curve  $C$  passing through  $p$  with multiplicity 1. So its strict transform intersects  $E$  transversely at one point, and  $\hat{C} \cdot E = 1$ . Since  $\hat{C} = \pi^*C - E$  by Lemma 3.1.6, and so  $(\pi^*C - E) \cdot E = 1$ . We know from the second equality that  $\pi^*C \cdot E = 0$ , so  $-E^2 = 1$ . Then we get  $E^2 = -1$ . □

Recall that a divisor  $D$  on a variety  $S$  is *ample* if some positive multiple of  $D$  defines an embedding of  $S$  into projective space. We now state the Hodge Index Theorem for surfaces.

**Theorem 3.1.8** (Hodge Index Theorem). *Let  $H$  be an ample divisor on a smooth surface  $S$ , and suppose  $D$  is a divisor,  $D \neq 0$ , with  $D.H = 0$ . Then  $D^2 < 0$ .*

The proof can be found in <sup>[Ha77]</sup>[Har77] Theorem V.1.9.

### 3.2 The Chow ring

<sup>FUL84</sup>  
[FUL84] Let  $X$  be an algebraic scheme. A  $k$ -cycle on  $X$  is a finite formal sum

$$\sum_i m_i [V_i]$$

where the  $V_i$  are  $k$ -dimensional subvarieties of  $X$ , and the  $m_i$  are integers. The group of  $k$ -cycles on  $X$ , denoted  $Z_k(X)$ , is the free Abelian group generated by the  $k$ -cycles. To a  $k$ -dimensional subvariety  $V$  of  $X$  corresponds  $[V]$  in  $Z_k(X)$ . The group of all cycles on  $X$  is denoted

$$Z(X) = \bigoplus_k Z_k(X),$$

and its elements are called *cycles*.

*Remark 3.2.1.*  $(n-1)$ -cycles are Weil divisors.

For any  $(k+1)$ -dimensional subvariety  $W$  of  $X$ , and any non-zero rational function  $f \in K(W)^\times$ , we can define a  $k$ -cycle  $[\operatorname{div}(f)]$  on  $X$  by

$$[\operatorname{div}(f)] = \sum_V \operatorname{ord}_V(f) [V]$$

where the sum is over all codimension-1 subvarieties  $V$  of  $W$ . Here,  $\operatorname{ord}_V(f)$  is the order function on  $K(W)^\times$  defined by the local ring  $\mathcal{O}_{W,V}$ .

A  $k$ -cycle  $\alpha$  is *rationaly equivalent to zero*, written  $\alpha \sim 0$ , if there are a finite number of  $(k+1)$ -dimensional subvarieties  $W_i$  of  $X$  and  $f_i \in K(W_i)^\times$  such that

$$\alpha = \sum_i [\operatorname{div}(f_i)].$$

Since  $[\operatorname{div}(f^{-1})] = -[\operatorname{div}(f)]$ , the cycles rationaly equivalent to zero form a subgroup of  $\operatorname{Rat}_k(X) \subseteq Z_k(X)$ . The quotient group

$$A_k(X) = Z_k(X) / \operatorname{Rat}_k(X)$$

is called the  $k$ -th *Chow group* of  $X$ , and its elements are called *Chow classes*. We say that two  $k$ -cycles are *rationaly equivalent* if they lie in the same class in  $A_k(X)$ .

**Definition 3.2.2.** The *Chow group* of  $X$  is the direct sum

$$A(X) = \bigoplus_k A_k(X).$$

An element  $\alpha \in A^k(X)$  is called a *cycle class*. We say that a cycle is *effective* if all its coefficients are non-negative. A cycle class is *effective* if it can be represented by an effective cycle.

Include example 1.3.1 from Fulton's book?

<sup>EH16</sup>  
[EH16] Theorem-definition 1.5 if we want to say that it's a ring.

### 3.2.1 Alternative definition of rational equivalence

Do we need this definition?

**Definition 3.2.3.** <sup>[EH16]</sup> Let  $\text{Rat}(X) \subseteq Z(X)$  be the subgroup generated by differences of the form

$$\langle \Phi \cap (\{t_0\} \times X) \rangle - \langle \Phi \cap (\{t_1\} \times X) \rangle$$

where  $t_0, t_1 \in \mathbb{P}^1$  and  $\Phi$  is a subvariety of  $\mathbb{P}^1 \times X$  not contained in any fiber  $\{t\} \times X$ . We say that two cycles are *rationally equivalent* if their difference lies in  $\text{Rat}(X)$ , and we say that two subschemes are rationally equivalent if their associated cycles are rationally equivalent.

Find a reference showing that this definition is equivalent to the previous one.

*Remark 3.2.4.* In many cases, it will be natural to study codimension  $r$  subvarieties of an  $n$ -dimensional variety  $X$ , and so we consider the  $(n - r)$ -th Chow group. We denote this as  $A^r(X) = A_{n-r}(X)$ . This group is called the *codimension  $r$  Chow group* of  $X$ .

For more details on the Chow ring, see <sup>[FUL84]</sup>[FUL84] chapter 1, or <sup>[EH16]</sup>[EH16] section 1.2.

### 3.2.2 The moving lemma

**Theorem 3.2.5** (The moving lemma). *Let  $X$  be a smooth quasi-projective variety.*

- (a) *For every  $\alpha, \beta \in A(X)$  there are generically transverse cycles  $A, B \in Z(X)$  with  $[A] = \alpha$  and  $[B] = \beta$ .*
- (b) *The class  $[A \cap B]$  is independent of the choice of such cycles  $A$  and  $B$ .*

We refer to <sup>[EH16]</sup>[EH16], Theorem 1.6 for a proof.

### 3.3 Chern classes

<sup>[EH16]</sup>  
[EH16] Recall that a Cartier divisor is defined through the vanishing loci of sections of line bundles. Chern classes are a way of generalizing this to vector bundles of higher rank, with the first Chern class corresponding to the Cartier divisor case.

Include assumptions on  $X$  to say this is an isomorphism.

#### 3.3.1 The first Chern class of a line bundle

<sup>[EH16]</sup>  
([EH16], 1.4.) Let  $\mathcal{L}$  be a line bundle on a variety  $X$  and  $\sigma$  be a rational section. Then on an open affine set  $U$  of a covering of  $X$ , we can write  $\sigma|_U = f_U/g_U$ , and define  $\text{Div}(\sigma)|_U = \text{Div}(f_U) - \text{Div}(g_U)$ . This definition agrees on the intersections of the open affine sets, and thus defines a Cartier divisor on  $X$ . Moreover, if  $\tau$  is another rational section of  $\mathcal{L}$ , then  $\alpha = \sigma/\tau$  is a well-defined rational function(!?), so  $\text{Div}(\sigma)$  and  $\text{Div}(\tau)$  are rationally equivalent divisors:

$$\text{Div}(\sigma) - \text{Div}(\tau) = \text{Div}(\alpha) \equiv 0 \text{ mod Rat}(X).$$

Thus, we can define the first Chern class of the line bundle  $\mathcal{L}$  as

$$c_1(\mathcal{L}) = [\text{Div}(\sigma)] \in A^1(X),$$

the rational equivalence class  $\text{Div}(\sigma)$  in the Chow group  $A^1(X)$ , for any nonzero rational section  $\sigma$ . This is well-defined, as it does not depend on the choice of rational section  $\sigma$ .

**Example 3.3.1.** A consequence is that  $c_1(\mathcal{O}_{\mathbb{P}^n}(d))$  is the class of any hypersurface of degree  $d$ . ▲

Improve or remove.

<sup>[EH16]</sup>  
**Proposition 3.3.2** ([EH16], Proposition 1.30). *If  $X$  is a variety of dimension  $n$ , then  $c_1$  is a group homomorphism*

$$c_1: \text{Pic}(X) \rightarrow A^1(X).$$

*If  $X$  is smooth, then  $c_1$  is an isomorphism.*

<sup>[EH16]</sup>  
[EH16] If  $Y \subseteq X$  is a divisor on a smooth variety  $X$ , then the ideal sheaf of  $Y$  is a line bundle denoted  $\mathcal{O}_X(-Y)$ , and its inverse in the Picard group is denoted  $\mathcal{O}_X(Y)$ . The inverse of the map  $c_1$  takes  $[Y]$  to  $\mathcal{O}_X(Y)$ .

#### 3.3.2 Chern classes of vector bundles

We now extend the definition of Chern classes to vector bundles of higher rank. Let  $\mathcal{E}$  be a vector bundle on an  $n$ -dimensional variety  $X$ , and let  $r = \text{rank}(\mathcal{E})$ . We follow the approach of <sup>[EH16]</sup>[EH16] section 5.2, giving an informal characterization rather than a formal definition.

In the case  $r = 1$ , the Chern class  $c_1(\mathcal{L})$  can be viewed as a measure of non-triviality. If  $c_1(\mathcal{L}) = 0$ , then  $\mathcal{L}$  has a nowhere-vanishing section, and  $\mathcal{L} \cong \mathcal{O}_X$ . We extend this idea to higher rank vector bundles by considering the “degeneracy locus” of collections of sections. Roughly, this is the locus where the sections become linearly dependent in the fibers of  $\mathcal{E}$ .

More precisely, the bundle  $\mathcal{E}$  is said to be trivial if and only if it has  $r$  global sections which are everywhere linearly independent. In this case, any set of  $r$  general sections



Necessary?

will do. We look at the exterior powers of  $\mathcal{E}$ . The  $(r-i)$ -th exterior power  $\bigwedge^{r-i} \mathcal{E}$  is a vector bundle of rank  $\binom{r}{r-i}$ , and general sections  $s_1, \dots, s_{r-i}$  induce sections of  $\bigwedge^{r-i} \mathcal{E}$  given by  $s_1 \wedge s_2 \wedge \dots \wedge s_{r-i}$ . For  $i = 0$ , we have a single section  $s_1 \wedge s_2 \wedge \dots \wedge s_r$  of the line bundle  $\bigwedge^r \mathcal{E}$ , and the zero locus is by definition  $c_1(\bigwedge^r \mathcal{E})$ . This is a class in  $A^1(X)$ , and we call it the *first Chern class* of  $\mathcal{E}$ .

In general, we have that for  $(r-i)$  general sections, we can consider the vanishing locus of

$$s_1 \wedge \dots \wedge s_{r-i} \in \bigwedge^{r-i} \mathcal{E}.$$

This vanishing locus is called the *i-th degeneracy locus* of the sections  $s_1, \dots, s_{r-i}$ .

### 3.3.3 Important results on Chern classes

Se kommentar om degeneracy loci.

**Theorem 3.3.3** ([EH16], Theorem 5.3 (a), (c)). *There is a unique way of assigning to each vector bundle  $\mathcal{E}$  on a smooth quasi-projective variety  $X$  a class  $c(\mathcal{E}) = 1 + c_1(\mathcal{E}) + \dots + c_n(\mathcal{E}) \in A(X)$  in such a way that*

- *If  $\mathcal{L}$  is a line bundle, then  $c(\mathcal{L})$  is  $1 + c_1(\mathcal{L})$ , where  $c_1(\mathcal{L}) \in A^1(X)$  is the class of the divisor of zeros minus the divisor of poles of any rational section of  $\mathcal{L}$ .*
- *If  $0 \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow 0$  is an exact sequence of vector bundles on  $X$ , then  $c(\mathcal{E}) = c(\mathcal{F})c(\mathcal{G}) \in A(X)$ . (Whitney sum formula!?)*

Se kommentar.

**Theorem 3.3.4** ([EH16], Theorem 5.12, Splitting principle). *Any identity(!?) among Chern classes of bundles that is true for bundles that are direct sums of line bundles is true in general.*

cor:sums

**Corollary 3.3.5** ([EH16], Corollary 5.5). *If  $\mathcal{E}$  is the direct sum of line bundles  $\mathcal{L}_i$ , then*

$$c(\mathcal{E}) = \prod c(\mathcal{L}_i) = \prod (1 + c_1(\mathcal{L}_i)).$$

Se veiledning fra 14.03.

The following examples are from [EH16], example 5.13, 5.14, and 5.16.

**Example 3.3.6** (Vanishing). If  $\mathcal{E}$  is a rank  $r$  vector bundle, then  $c_i(\mathcal{E}) = 0$  for  $i > r$ . To see why, apply the splitting principle to write  $\mathcal{E} = \bigoplus_{i=1}^r \mathcal{L}_i$  for line bundles  $\mathcal{L}_i$ . Then, since  $c(\mathcal{L}_i) = 1 + c_1(\mathcal{L}_i)$ , the total Chern class  $c(\mathcal{E}) = \prod (1 + c_1(\mathcal{L}_i))$  by Corollary 3.3.5. This is a product of  $r$  factors each of degree at most 1, so any Chern class of degree greater than  $r$  must be zero. ▲

**Example 3.3.7** (Duals). If  $\mathcal{E} = \bigoplus \mathcal{L}_i$ , Then

$$c(\mathcal{E}^\vee) = \prod c(\mathcal{L}_i^\vee) = \prod (1 - c_1(\mathcal{L}_i))$$

since  $c_1(\mathcal{L}^\vee) = -c_1(\mathcal{L})$  when  $\mathcal{L}$  is a line bundle. Given this, Whitney's formula gives

$$ci(\mathcal{E}^\vee) = (-1)^i ci(\mathcal{E})$$

▲

**Example 3.3.8** (Symmetric squares). Suppose  $\mathcal{E}$  is a bundle of rank 2. If  $\mathcal{E}$  splits as a direct sum  $\mathcal{L} \oplus \mathcal{M}$  of line bundles  $\mathcal{L}$  and  $\mathcal{M}$  with Chern classes  $c_1(\mathcal{L}) = \alpha$ , and  $c_1(\mathcal{M}) = \beta$ , then by Whitney's formula.

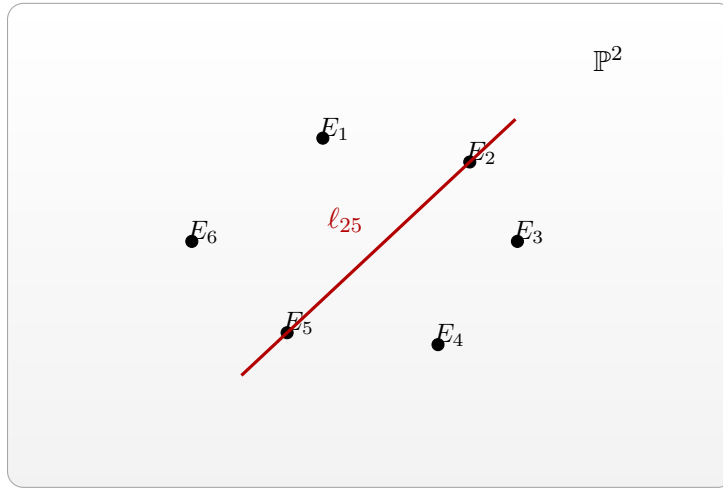
Finish.



Consider tensor products.

**Proposition 3.3.9** (<sup>EH16</sup>[EH16], Proposition 6.4). *Let  $V$  be an  $(n+1)$ -dimensional vector space, and let  $S \subseteq V \otimes \mathcal{O}_{\mathcal{G}}$  (Introduce this earlier!?) be the tautological rank- $(k+1)$  subbundle on the Grassmannian  $\mathcal{G} = \mathcal{G}(k, \mathbb{P}V)$  of  $k$ -planes in  $\mathbb{P}V \cong \mathbb{P}^n$ . A form  $g$  of degree  $d$  on  $\mathbb{P}V$  gives rise to a global section  $\sigma_g$  of  $\text{Sym}^d S^\vee$  whose zero locus is the Fano scheme  $F_k(X)$ , where  $X = V(g) \subseteq \mathbb{P}V$  is the hypersurface defined by  $g$ . Thus, when  $F_k(X)$  has expected codimension  $\binom{k+d}{k} = \text{rank}(\text{Sym}(S^\vee))$  in  $\mathcal{G}$ , its class in the Chow ring of  $\mathcal{G}$  is given by the top Chern class*

$$\deg F_k(X) = c_{\binom{k+d}{k}}(\text{Sym}^d S^\vee) \in A(\mathcal{G}).$$



Highlighted: strict transform of the line through  $E_2$  and  $E_5$  in  $\mathbb{P}^2$ .

Figure 3.1: The projective plane  $\mathbb{P}^2$  with six marked blow-up points and a strict transform of the line  $\ell_{25}$ .

fig:blowup6

### 3.4 27 lines on a cubic surface

From Corollary 2.6.6, we know that a generic smooth cubic surface contains finitely many lines. We will now show that it contains exactly 27 lines. The following is the main theorem of this section.

thm:27

**Theorem 3.4.1.** *Each smooth cubic surface in  $\mathbb{P}^3$  contains exactly 27 distinct(!?) lines.*

This will be the main theorem of this section. We will prove it in two different ways. The first will be a well known proof, relating cubic surfaces to blow-ups of  $\mathbb{P}^2$  in 6 points. The second proof will use intersection theory on the Grassmannian  $\mathbb{G}(1,3)$ . We begin with the first proof.

Describe second result better.

#### 3.4.1 Cubic surfaces as blow-ups of $\mathbb{P}^2$ in 6 points

Make a better figure of the blow-up of  $\mathbb{P}^2$  in 6 points.

This proof will rely on theory which is not in the scope of this thesis, but it will allow us to draw a picture to show a representation of the 27 lines themselves, which we will use later.

First, the points  $P_1, \dots, P_6$  in  $\mathbb{P}^2$  are said to be in *general position* if no three of the points are collinear, and no six lie on a conic.

Se kommentar.

prop:blowup6

**Proposition 3.4.2** ([<sup>Ha77</sup>Har77], 4.7/4.7.3). *Let  $X \subset \mathbb{P}^3$  be a smooth cubic surface. Then there exist points  $P_1, \dots, P_6 \in \mathbb{P}^2$  in general position such that  $X$  is isomorphic to the blow-up of  $\mathbb{P}^2$  in these points.*

This is part of a bigger class of surfaces, those being the *Del Pezzo* surfaces. These are surfaces which (Consider continuing, and maybe cite Beauville if so!?)

prop:27

**Proposition 3.4.3.** (<sup>Ha77</sup>[Har77], 4.9) *Let  $X \subset \mathbb{P}^3$  be a smooth cubic surface. Then  $X$  contains exactly 27 lines. Each has self-intersection  $-1$ , and they are the only irreducible curves with negative self-intersection. They are:*

- *The exceptional curves  $E_1, \dots, E_6$  (6 of these)*
- *The strict transform of lines in  $\mathbb{P}^2$  containing pairs of points  $P_i, P_j$ ,  $i \neq j$ , denoted  $\ell_{ij}$  (15 of these).*
- *The strict transform of conics in  $\mathbb{P}^2$  not containing  $P_i$  denoted  $C_i$  (6 of these).*

$\mathbb{P}^2$  blown up in 6 points is represented in Fig. 3.1.

### 3.4.2 Cayley-Salmon (!?)

Decide if we want to include this classic proof. We could do as EO, finding 27 lines on the Fermat cubic.

**Lemma 3.4.4.** *The Fermat cubic surface  $X \subset \mathbb{P}^3$  defined by the vanishing of the form  $F = x_0^3 + x_1^3 + x_2^3 + x_3^3$  contains exactly 27 distinct lines.*

### 3.4.3 Proof by degree of Chern classes

We will now give a proof using intersection theory on the Grassmannian  $\mathbb{G}(1, 3)$ . Recall from Example 2.2.4 that  $\mathbb{G}(1, 3)$  has dimension 4.

*Proof of Theorem 3.4.1.* Given a smooth cubic surface  $X \subset \mathbb{P}^3$  determined by the vanishing of a cubic form  $F$  in four variables, we wish to determine the degree of the locus in  $\mathbb{G}(1, 3)$  of lines contained in  $X$ .

Observe that if we fix a line  $L$  in  $\mathbb{P}^3$ , then the condition that  $L$  lies in  $X$  is equivalent to the condition that the restriction of  $F$  to  $L$  is identically zero. Note that the restriction map from the 20-dimensional vector space of cubic forms on  $\mathbb{P}^3$  to the 4-dimensional vector space of cubic forms on  $L$ ,  $V_L = H^0(L, \mathcal{O}_L(3))$  is a linear surjection, and the condition  $L \subseteq X$  is that  $F$  maps to 0 in  $V_L$ . This imposes four linear conditions on the coefficients of  $F$  (!?).

As the line varies over  $\mathbb{G}(1, 3)$ , the vector spaces  $V_L$  fit together to form a vector bundle of rank 4,  $\mathcal{V}$  on  $\mathbb{G}(1, 3)$ , with fiber over the point  $[L]$  being  $V_L$ . A cubic form  $F$  on  $\mathbb{P}^3$ , through its restriction to each  $V_L$ , defines a global section  $\sigma_F$  of this vector bundle. Thus, the locus of lines contained in the cubic surface  $X$  is the zero locus of the section  $\sigma_F$ . Assuming that this zero locus is zero-dimensional (!?), its class in  $A(\mathbb{G}(1, 3))$  is given by the top Chern class  $c_4(\mathcal{V})$  of the vector bundle  $\mathcal{V}$ . The number of lines contained in  $X$  is therefore the degree of this class (!?).

To compute this degree, we first express  $\mathcal{V}$  in terms of the universal subbundle  $\mathcal{S}$  on  $\mathbb{G}(1, 3)$ . Recall that the fiber of  $\mathcal{S}$  over a point  $[L]$  is the 2-dimensional vector space corresponding to the line  $L \subseteq \mathbb{P}^3$ . The restriction of cubic forms on  $\mathbb{P}^3$  to  $L$  factors through the symmetric third power of this vector space, so we have an isomorphism of vector bundles

$$\mathcal{V} \cong \text{Sym}^3(\mathcal{S}^\vee).$$

Using the splitting principle, we can compute the Chern classes of  $\mathcal{V}$  in terms of the Chern classes of  $\mathcal{S}^\vee$ . Let the Chern roots of  $\mathcal{S}^\vee$  be  $\alpha_1$  and  $\alpha_2$ . Then the Chern roots of  $\mathcal{V}$  are given by

$$3\alpha_1, 2\alpha_1 + \alpha_2, \alpha_1 + 2\alpha_2, 3\alpha_2.$$

Thus, the top Chern class of  $\mathcal{V}$  is

$$\begin{aligned} c_4(\mathcal{V}) &= (3\alpha_1)(2\alpha_1 + \alpha_2)(\alpha_1 + 2\alpha_2)(3\alpha_2) \\ &= 9\alpha_1\alpha_2(2\alpha_1^2 + 5\alpha_1\alpha_2 + 2\alpha_2^2) \\ &= 9c_2(\mathcal{S}^\vee)(2c_1(\mathcal{S}^\vee)^2 + 5c_2(\mathcal{S}^\vee)). \end{aligned}$$

To compute the degree of this class, we express it in terms of the Schubert classes  $\sigma_1 = c_1(\mathcal{S}^\vee)$  and  $\sigma_{1,1} = c_2(\mathcal{S}^\vee)$ , and use the intersection numbers computed in (!?). Therefore, a smooth cubic surface in  $\mathbb{P}^3$  contains exactly 27 distinct lines.  $\square$

Finish proof properly.

Legg til figur av Clebsch.



## Chapter 4

# Configurations of lines

configs

### 4.1 Surfaces including m-crosses

In this section we will study the problem of finding a smooth hypersurface  $X_d \subset \mathbb{P}^3$  of degree  $d$ , containing an  $m$ -cross.

**Definition 4.1.1.** An  $m$ -cross is a union of  $m$  distinct lines  $l_1, l_2, \dots, l_m$  in  $\mathbb{P}^3$  which intersect at a common point  $p$  and lying in a common plane  $H$ . We will write  $Z_m = \bigcup_{i=1}^m l_i$ .

We require the lines to lie in a plane because if this is not true, then the point  $p$  gives rise to a tangent space of dimension 3, which cannot be included in a surface in  $\mathbb{P}^3$ .

Gjør dette argumentet rigorøst. Se notat fra 6.11.

It is worth noting the specific question we are asking, which is that given an  $m$ -cross, does there exist a smooth hypersurface of degree  $d$  including it. We do not ask whether an  $m$ -cross is to be expected inside a general hypersurface.

Se kommentar.

rem:autPm

**Remark 4.1.2.** Note that for automorphisms of  $\mathbb{P}^3$ , lines are sent to lines. Therefore, we can study the lines  $l_1 = (\alpha : \beta : 0 : 0)$ ,  $l_2 = (\alpha : 0 : \gamma : 0)$ ,  $l_3 = (\alpha : \delta : \lambda_3 \delta : 0)$ ,  $\dots$ ,  $l_m = (\alpha : \delta : \lambda_m \delta : 0)$  for  $\lambda_i \in \mathbb{N}$ , where  $\alpha, \beta, \gamma$ , and  $\lambda_i$  are parameters which vary, by taking the automorphism of  $\mathbb{P}^3$  which sends these lines to the given lines, and then study the hypersurface of degree  $d$  in  $\mathbb{P}^3$  which contains these lines. Note also that

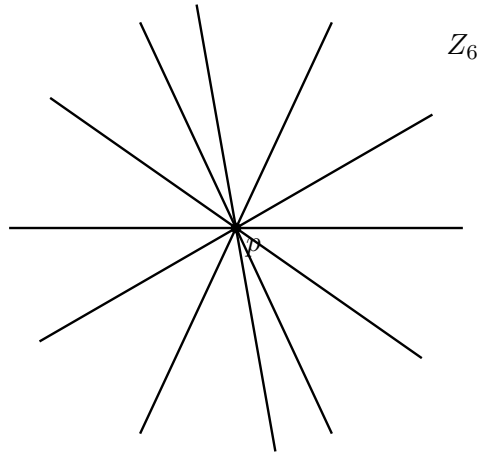


Figure 4.1: An example of a 6-cross in  $\mathbb{P}^3$ .

fig:Z\_6

$l_1 = (1 : 0 : 0 : 0)$  since we are working in projective space. This automorphism will also take a smooth hypersurface of degree  $d$  to a smooth hypersurface of degree  $d$ , so showing properties for lines in this form is sufficient for the general case. In this way, we can assume without loss of generality that the plane  $H$  containing the lines is given as  $H = V(x_3)$ , and we notice that a line  $l_i = V(H) \cap V(g_i)$  for some  $g_i \in \mathbb{C}[x_0, x_1, x_2, x_3]$ . The lines given above can then be described as the zero loci of the following polynomials:

$$l_1 = V(x_3, x_2), l_2 = V(x_3, x_1), l_3 = V(x_3, x_1 - \lambda_3 x_2), \dots, l_m = V(x_3, x_1 - \lambda_m x_2). \quad (4.1)$$

eq:1i

We are interested in both the existence and non-existence of degree  $d$  hypersurfaces containing  $m$ -crosses. We first consider surfaces with degree less than  $k$ , but more than 1 (degree 1 provides us a simple choice of surface including an  $m$ -cross, namely  $H$ ).

lem:H

**Lemma 4.1.3.** *If  $S$  is a smooth surface in  $\mathbb{P}^3$  of degree  $d > 1$ , it does not contain a plane.*

*Proof.* Assume for contradiction that  $S = V(g)$  contains a plane  $H$ . We let  $H = V(x_3)$  without loss of generality (refer to a lemma or previous discussion?)(!). Then  $x_3$  divides  $g$ , and we write  $g = x_3 h$  for some  $h \in \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1}$ . We then use the Jacobi criterion (15.12 in EO!) on  $x_3 h$ , and get that  $S$  is singular on the whole of  $V(x_3) \cap V(h)$ . This is non-empty by Bezout's theorem (Theorem 1.3.2), giving a contradiction. (Use  $d > 1$  to see that  $\deg(h) = d - 1 > 0$ )(!).  $\square$

Dette må forbedres.

**Lemma 4.1.4.** *Given an  $m$ -cross, there does not exist any smooth hypersurface of degree  $1 < d < m$  such that the lines are contained in the hypersurface.*

*Proof.* Assume for contradiction that there exists a smooth hypersurface of degree  $1 < d < m$  such that the lines are contained in the hypersurface. Then we can fix another line  $\ell$  in  $H$ , which does not intersect  $p$ , and which does not lie in the hypersurface  $X_d$  by Lemma 4.1.3.  $\ell$  will intersect each of the lines at a point  $p_i$ , all being distinct since the lines only intersect in  $p$ .  $\ell$  will also intersect  $X_d$  at the points  $p_i$ , thus intersecting the hypersurface at  $m$  points. By Bezout's theorem (Theorem 1.3.2), the intersection of a line with a hypersurface of degree  $d$  is at most  $d$  points, giving a contradiction.  $\square$

We now move on to the case of existence, with the following proposition:

**Proposition 4.1.5.** *For a given  $m$ -cross, we can find a smooth hypersurface of degree  $m$  such that the lines are contained in the hypersurface.*

*Proof.* Let  $\mathfrak{L}$  be the linear system of hypersurfaces of degree  $m$  in  $\mathbb{P}^3$  containing the lines  $l_1, \dots, l_m$  (Prove this is not empty, because of singular surfaces.)(!). Recall that Bertini's theorem (Theorem 1.3.1) tells us that the general member of  $\mathfrak{L}$  is smooth outside the base locus of  $\mathfrak{L}$ . We can therefore narrow our search of singularities to the base locus. The base locus of  $\mathfrak{L}$  is the union of the lines  $l_1, \dots, l_m$ . To see why, consider a point  $x \in \mathbb{P}^3$  which does not lie on one of the lines. We want to show that for each of these points, we can find a surface which excludes the points, but includes the surface. First, if the point is not in the plane  $H$ , we can exclude it by the plane itself, so we assume the point lies in  $H$ , but not in one of the lines. Then we consider the loci which define the lines, as discussed in Remark 4.1.2, denoted  $l_i = V(H) \cap V(g_i)$ . Then all the lines  $l_1, \dots, l_m$  are included in  $V(\prod_{i=1}^m g_i)$ , while  $x$  is not.



Since we know exactly the base locus, we want to find a function  $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_m$  such that  $V(f)$  is smooth on the lines  $l_1, \dots, l_m$ . Bertini's theorem then gives us that we can find a hypersurface which is also smooth outside the base locus of  $\mathfrak{L}$ .

Now we take inspiration from [eq:1.1](#), and study the function  $f = g(x_1, x_2) + h(x_0, x_3)$ , where  $h = x_0^{m-1}x_3$  and  $g = \prod_{i=3}^m l_i = x_1x_2 \prod_{i=3}^m (x_1 - \lambda_i x_2)$ , giving the following polynomial:

$$f(x_0, x_1, x_2, x_3) = x_1x_2 \prod_{i=3}^m (x_1 - \lambda_i x_2) + x_0^{m-1}x_3.$$

We can see that  $f$  is a polynomial of degree  $m$ , and that  $V(f)$  includes the lines  $l_1, \dots, l_m$ , since  $f(\alpha : \beta : 0 : 0) = f(\alpha : 0 : \gamma : 0) = f(\alpha : \delta : \lambda_i \delta : 0) = 0$ . We must then check that  $f$  is smooth on the base locus of  $\mathfrak{L}$ , which is to check that  $\bigcap_{i=0}^3 (\partial_i f = 0) = \emptyset$ .

The derivatives are given as follows:

$$\begin{aligned} \partial_0 f &= \partial_0 g + \partial_0 h = 0 + (m-1)x_0^{m-2}x_3 = (m-1)x_0^{m-2}x_3, \\ \partial_1 f &= \partial_1 g + \partial_1 h = x_2 \prod_{i=3}^m (x_1 - \lambda_i x_2) + x_1x_2 \sum_{i=3}^m \prod_{\substack{j=3 \\ j \neq i}}^m (x_1 - \lambda_j x_2), \\ \partial_2 f &= \partial_2 g + \partial_2 h = x_1 \prod_{i=3}^m (x_1 - \lambda_i x_2) + x_1x_2 \sum_{i=3}^m (-\lambda_i) \cdot \prod_{\substack{j=3 \\ j \neq i}}^m (x_1 - \lambda_j x_2), \\ \partial_3 f &= \partial_3 g + \partial_3 h = 0 + x_0^{m-1} = x_0^{m-1}. \end{aligned}$$

We then find that on the base locus of  $\mathfrak{L}$ , the following hold:

- $\partial_0 f = 0$  on the entire base locus, since it contains the factor  $x_3$ , which is 0 on the entire plane.
- $\partial_1 f = 0$  on the line  $l_0 = (\alpha : \beta : 0 : 0)$ , since  $\partial_1 f$  contains the factor  $x_2$ , and is non-zero elsewhere on the base locus.
- $\partial_2 f = 0$  on the line  $l_1 = (\alpha : 0 : \gamma : 0)$ , since  $\partial_2 f$  contains the factor  $x_1$ , and is non-zero elsewhere on the base locus.
- $\partial_3 f = 0$  if and only if  $x_0 = 0$ , which is the case for the following points:  $(0 : \beta : 0 : 0), (0 : 0 : \gamma : 0), (0 : \delta : \lambda_i \delta : 0)$  for  $3 \leq i \leq m$ .

We can see that the intersection of all of these zero loci is empty, showing that the function  $f$  is smooth on the base locus of  $\mathfrak{L}$ . Thus, we can find a smooth hypersurface of degree  $k$  such that the lines are contained in the hypersurface.  $\square$

Fiks bevis. Se kommentar.

By the results above, we get the following corollary:

**Corollary 4.1.6.** *Given an m-cross, we can find a smooth hypersurface of degree  $d$  such that the lines are contained in the hypersurface if and only if  $d \geq m$  or  $d = 1$ .*

*Proof.* If we have  $m$  lines, we can find a smooth hypersurface of degree  $d \geq m + 1$  by including arbitrary lines  $l_{m+1}, l_{m+2}, \dots, l_d$  lying in the plane. For the case  $d = 1$ , we can take the plane  $H$  containing the lines.  $\square$

## 4.2 The set of degree $d$ surfaces including $m$ -crosses

In the previous section we studied the existence and non-existence of smooth hypersurfaces of degree  $d$  including an  $m$ -cross. We will now study the dimension of the set of degree  $d$  surfaces including  $m$ -crosses. Let  $Z_m$  be an  $m$ -cross lying in a plane  $H$ , which we will assume is the plane  $V(x_3)$  by an automorphism of  $\mathbb{P}^3$ .

**Proposition 4.2.1.** *Let  $A_m = \{S_d \subseteq \mathbb{P}^3 \mid Z_m \subset S_d\} \subseteq \{S_d \subseteq \mathbb{P}^3\}$  be the set of degree  $d$  surfaces containing a general  $Z_m$ . Then  $\dim A_m = \binom{d-m+2}{2} + \binom{d+2}{3} - 1$ .*

*Proof.* Let  $S_d = V(f)$  for some  $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_d$  be a surface which includes  $Z_m$ . Since  $Z_m$  is included in  $H$ , we write  $f$  as  $f = g(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$ , separating the part of  $f$ ,  $x_3 h(x_0, x_1, x_2, x_3)$ , which vanishes on the plane. Saying that  $Z_m \subseteq S_d$  is the same as saying  $f|_{Z_m} = 0$ , which is equivalent to  $g(x_0, x_1, x_2)|_{Z_m} = 0$ , since  $x_3 h(x_0, x_1, x_2, x_3)|_{Z_m}$  already vanishes. This is equivalent to  $l_1 l_2 \cdots l_m$  divides  $g(x_0, x_1, x_2)$ , meaning  $g(x_0, x_1, x_2) = l_1 l_2 \cdots l_m \cdot j(x_0, x_1, x_2)$  for some  $j(x_0, x_1, x_2) \in \mathbb{C}[x_0, x_1, x_2]_{d-m}$ . Thus, we can write  $f$  as  $f = l_1 l_2 \cdots l_m \cdot j(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$ , where  $h(x_0, x_1, x_2, x_3) \in \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1}$ .

Since we can choose  $j(x_0, x_1, x_2)$  and  $h(x_0, x_1, x_2, x_3)$  freely, we get that  $\dim A_m = \dim \mathbb{C}[x_0, x_1, x_2]_{d-m} + \dim \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1} - 1 = \binom{d-m+2}{2} + \binom{d+2}{3} - 1$ . We subtract 1 to account for the fact that we are working in projective space.  $\square$

Include examples?

## 4.3 Surfaces including $m$ -gons

In this section, we will consider  $m$ -gons in  $\mathbb{P}^3$  and their relationship with hypersurfaces.

**Definition 4.3.1.** An  $m$ -gon is a union of  $m$  lines  $l_1, l_2, \dots, l_m$  such that  $l_i$  intersects  $l_{i+1}$  for  $1 \leq i < m$ , and  $l_m$  intersects  $l_1$ . We will write  $C_m = \bigcup_{i=1}^m l_i$  and denote the intersection points as  $p_i = l_i \cap l_{i+1}$  for  $1 \leq i < m$ , and  $p_m = l_m \cap l_1$ .

**Definition 4.3.2.** A *regular*  $m$ -gon is an  $m$ -gon such that  $l_i \neq l_j$  for  $i \neq j$ , and no lines other than the adjacent ones intersect.

The notion of a regular  $m$ -gon is important. For example, a non-regular  $m$ -gon can be  $m$  lines all intersecting at a single point, i.e. an  $m$ -cross. Such cases would not hold the same properties as what one would expect from  $m$ -gons such as the one drawn in Fig. 4.2. Next, we consider the set of  $m$ -gons for fixed  $m$ .

**Definition 4.3.3.** Let  $\mathcal{C}_m \subseteq \mathbb{G}(1, 3)^m$  be the family of all  $m$ -gons in  $\mathbb{P}^3$ .

We can describe  $\mathcal{C}_m$  as the set

$$\mathcal{C}_m = \{(l_1, l_2, \dots, l_m) \in \mathbb{G}(1, 3)^m \mid l_i \cap l_{i+1} \neq \emptyset \text{ for } 1 \leq i < m, \text{ and } l_m \cap l_1 \neq \emptyset\}.$$

Now we aim to show that  $\mathcal{C}_m$  is an irreducible variety, and compute its dimension. To do this, we first define a related variety which we will study further in Section 4.4, that being  $m$ -chains, which are configurations of lines much like  $m$ -gons, only without the requirement that  $l_m$  shall intersect  $l_1$ . In this way,  $m$ -gons can be thought of as being iteratively constructed by adding one line at a time, each time forming an  $i$ -chain for  $i = 1, \dots, m-1$ , until we finally add the last line which intersects the first line, closing the loop. This construction allows us to first study the set of  $m$ -chains, and then viewing the projection map of  $\mathcal{C}_m$  onto  $\mathcal{Q}_m$  to get the desired results for  $m$ -gons.

**Definition 4.3.4.** We define  $\mathcal{Q}_m \subseteq \mathbb{G}(1, 3)^m$  to be the subset of  $m$ -tuples of lines  $(l_1, \dots, l_m)$  such that they form an  $m$ -chain.

First we need the following lemma about Schubert varieties.

em:Schubert

**Lemma 4.3.5.** *The Schubert variety  $\Sigma_L = \{l \in \mathbb{G}(1, 3) \mid l \cap L \neq \emptyset\}$  for a fixed line  $L$  is irreducible of dimension 3.*

Put this in background chapter on Schubert cells, if we include this.

*Proof.* <sup>[EH16]</sup>[EH16], section 3.3.1. First, consider the incidence correspondence  $\Gamma = \{(l, p) \in \mathbb{G}(1, 3) \times L \mid p \in l\}$ . The projection map  $g: \Gamma \rightarrow \mathbb{G}(1, 3)$  given by  $g(l, p) = l$  gives us that  $\Sigma_L$  is the image of  $g$ . Next, the fiber of the projection map  $f: \Gamma \rightarrow L$  given by  $f(l, p) = p$  is the set of lines through a point  $p \in L$ , meaning the fiber is isomorphic to  $\mathbb{P}^2$ . Since  $L$  is isomorphic to  $\mathbb{P}^1$ , it is irreducible of dimension 1. Since  $f$  is proper, and all the fibers are irreducible of dimension 2, we have from Lemma 1.3.4 that  $\Gamma$  is irreducible of dimension  $2 + 1 = 3$ .  $\square$

Se kommentar.

lem:dimQm

**Lemma 4.3.6.**  $\mathcal{Q}_m$  is an irreducible variety of dimension  $3m + 1$  for  $m \geq 1$ .

*Proof.* We first observe that  $\mathcal{Q}_1 = \mathbb{G}(1, 3)$ , which is irreducible of dimension 4. We then proceed by induction on  $m$ , assuming that  $\mathcal{Q}_m$  is irreducible of dimension  $M$  for some integer  $M$ .

Let  $m \geq 2$ , and consider the projection map  $\pi_m: \mathcal{Q}_m \rightarrow \mathcal{Q}_{m-1}$  given by  $\pi(l_1, \dots, l_m) = (l_1, \dots, l_{m-1})$ . The fiber over a point is given by  $\pi^{-1}(l_1, \dots, l_{m-1}) = \{(l_1, \dots, l_{m-1}, l_m) \in \mathcal{Q}_m \mid l_m \cap l_{m-1} \neq \emptyset\}$  which is isomorphic to  $\Sigma_{l_{m-1}} = \{l \in \mathbb{G}(1, 3) \mid l \cap l_{m-1} \neq \emptyset\}$ . From Lemma 4.3.5 we know that  $\Sigma_{l_{m-1}}$  is irreducible of dimension 3, and by the induction hypothesis that  $\mathcal{Q}_{m-1}$  is irreducible of dimension  $M$ . Then, since  $\pi_m$  is proper, with irreducible fiber of dimension 3, we again have that  $\mathcal{Q}_m$  is irreducible of dimension  $M + 3$  by Lemma 1.3.4.  $M = 4$  for  $m = 1$ , and so we get that  $\dim \mathcal{Q}_m = 4 + 3(m - 1) = 3m + 1$ .  $\square$

**Lemma 4.3.7.**  $\Sigma_L \cap \Sigma_{L'} = \{l \in \mathbb{G}(1, 3) \mid l \cap L \neq \emptyset, l \cap L' \neq \emptyset\}$ , for distinct non-intersecting lines  $L, L'$ , is irreducible of dimension 2.

*Proof.* Consider the incidence correspondence

$$\Gamma = \{(l, p, p') \in \mathbb{G}(1, 3) \times L \times L' \mid p \in l, p' \in l\}.$$

The projection map  $g: \Gamma \rightarrow \mathbb{G}(1, 3)$  given by  $g(l, p, p') = l$  has image  $\Sigma_L \cap \Sigma_{L'}$ . Next, the fiber of the projection map  $f: \Gamma \rightarrow L \times L'$  given by  $f(l, p, p') = (p, p')$  is the set of lines through two distinct points  $p \in L$  and  $p' \in L'$ , which is a single line. Since  $L \times L' \simeq \mathbb{P}^1 \times \mathbb{P}^1$  is irreducible of dimension 2, and all fibers of  $f$  are irreducible of dimension 0, we have from Lemma 1.3.4 that  $\Gamma$  is irreducible of dimension  $2 + 0 = 2$ . Thus, the image  $\Sigma_L \cap \Sigma_{L'}$  is also irreducible of dimension 2.  $\square$

**Corollary 4.3.8.**  $\mathcal{C}_m$  is an irreducible variety of dimension  $3m$  for  $m \geq 3$ .

Se over.

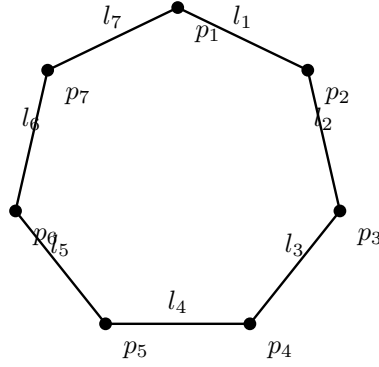


Figure 4.2: 7-gon drawn in a plane; each vertex is an intersection point  $p_i$  and the line segments represent the lines  $l_i$ . Note that the lines in reality extend infinitely in both directions.

fig:m-gon

*Proof.* Consider the projection map  $\pi: \mathcal{C}_m \rightarrow \mathcal{Q}_{m-1}$  given by  $\pi(l_1, \dots, l_m) = (l_1, \dots, l_{m-1})$ . The fiber over a point is given by

$$\pi^{-1}(l_1, \dots, l_{m-1}) = \{(l_1, \dots, l_m) \in \mathcal{C}_m \mid l_m \cap l_1 \neq \emptyset, l_m \cap l_{m-1} \neq \emptyset\},$$

which is isomorphic to  $\Sigma_{l_1} \cap \Sigma_{l_{m-1}} = \{l \in \mathbb{G}(1, 3) \mid l \cap l_1 \neq \emptyset, l \cap l_{m-1} \neq \emptyset\}$ . From the previous lemma, we know that this is irreducible of dimension 2, and from Lemma 4.3.6 that  $\mathcal{Q}_m$  is irreducible of dimension  $3m + 1$ . Since  $\pi$  is proper, with irreducible fiber of dimension 2, we have from Lemma 1.3.4 that  $\mathcal{C}_m$  is irreducible of dimension  $3(m - 1) + 1 + 2 = 3m$ .  $\square$

lem:gon-open

**Lemma 4.3.9.** *The set of regular  $m$ -gons is an open subset of  $\mathcal{C}_m$ .*

*Proof.* Let  $R_m \subset \mathcal{C}_m$  be the locus of regular  $m$ -gons. By definition a tuple  $(l_1, \dots, l_m) \in \mathcal{C}_m$  is regular if and only if: (i)  $l_i \neq l_j$  for all  $i \neq j$ ; (ii)  $l_i \cap l_j = \emptyset$  whenever  $j \neq i \pm 1 \pmod{m}$  (i.e. only adjacent pairs meet).

Condition (i) can be described by the locus of a finite number of diagonals  $\Delta_{ij} = \{(l_1, \dots, l_m) \mid l_i = l_j\}$  in  $\mathbb{G}(1, 3)^m$ , and so its complement is open. Condition (ii) is described by the complements of a finite number of zero loci of the equations  $l_i \cap l_j \neq \emptyset$  for adjacent pairs  $(i, j)$ , and so this condition is also open.  $\square$

Another lemma we will need is the following.

lem:h00c

**Lemma 4.3.10.** *The global sections of the structure sheaf of a regular  $m$ -gon  $C_m$  has dimension  $h^0(C_m, \mathcal{O}_{C_m}(d)) = md$ , for  $d \geq 1$ .*

*Proof.* First, consider the short exact sequence:

$$0 \rightarrow \mathcal{O}_{C_m}(d) \rightarrow \bigoplus_{i=1}^m \mathcal{O}_{l_i}(d) \rightarrow \bigoplus_{i=1}^m \mathcal{O}_{p_i} \rightarrow 0.$$

This sequence comes from the fact that sections on  $C_m$  can be viewed as collections of sections on each line  $l_i$ , which agree at the intersection points  $p_i$ , so we note that this argument requires the regularity of  $C_m$ . Taking the long exact sequence in cohomology, we get

$$0 \rightarrow H^0(C_m, \mathcal{O}_{C_m}(d)) \rightarrow H^0\left(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)\right) \rightarrow H^0\left(\bigoplus_{i=1}^m \mathcal{O}_{p_i}\right) \rightarrow H^1(C_m, \mathcal{O}_{C_m}(d)) \rightarrow 0,$$

Er kohomologi-gruppene riktige her? Skal jeg heller skrive  $H^0(C_m, \bigoplus \mathcal{O}_{l_i}(d))$  f.eks?

since  $H^1(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) = 0$  for all  $d \geq 0$ . Taking dimensions, we get

$$h^0(C_m, \mathcal{O}_{C_m}(d)) = h^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) + h^1(C_m, \mathcal{O}_{C_m}(d)) - h^0(\bigoplus_{i=1}^m \mathcal{O}_{p_i}).$$

Since  $h^0(\bigoplus_{i=1}^m \mathcal{O}_{p_i}) = m$ , and  $h^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) = m \cdot \binom{d+1}{1}$ , this gives us  $h^0(C_m, \mathcal{O}_{C_m}) = md + h^1(C_m, \mathcal{O}_{C_m}(d))$ . We want to show that  $h^1(C_m, \mathcal{O}_{C_m}(d)) = 0$ . To see this, we show that the map from  $H^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d))$  to  $H^0(\bigoplus_{i=1}^m \mathcal{O}_{p_i})$  is surjective. As discussed, the map  $\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d) \rightarrow \bigoplus_{i=1}^m \mathcal{O}_{p_i}$  takes sections  $(f_{12}, f_{23}, \dots, f_{m1})$  to  $(f_{12}(p_1) - f_{m1}(p_1), f_{23}(p_2) - f_{12}(p_2), \dots, f_{m1}(p_m) - f_{(m-1)m}(p_m))$ . Our goal then is to show that we can send these differences to a basis for  $\bigoplus_{i=1}^m \mathcal{O}_{p_i}$ . We will show that the element  $(1, 0, \dots, 0)$  is in the image of our map. That is, we are looking for a collection of sections  $(f_{12}, f_{23}, \dots, f_{m1})$  which agree on all points except  $p_1$ . Finding sections which agree on points other than  $p_1$  is as simple as letting them be the zero sections, which can all be considered to have arbitrary degrees. Therefore, let  $f_{23} = \dots = f_{m1} = 0$ . Then we need to find a section  $f_{12}$  which evaluates to 1 at  $p_1$  and 0 at  $p_2$ . Let

$$f'_{12} = \left(\frac{x_0}{\alpha}\right)^d + \left(\frac{x_1}{\beta}\right)^d + \left(\frac{x_2}{\gamma}\right)^d - 3\left(\frac{x_3}{\delta}\right)^d,$$

where  $p_2 = (\alpha : \beta : \gamma : \delta)$ . This is a degree  $d$  section on the line  $l_1$ , and evaluates to 0 at  $p_2$ , and a nonzero number  $\lambda$  at  $p_1$ . Then we can scale this section by

$$f_{12} = \frac{f'_{12}}{\lambda},$$

which then evaluates to 1 at  $p_1$  and 0 at  $p_2$ .

Forbedre argument. Kanskje bruk figur av 7-gon.

Thus, we have found sections  $(f_{12}, f_{23}, \dots, f_{m1})$  which map to  $(1, 0, \dots, 0)$ . By similar reasoning, we can find sections mapping to any basis element of  $\bigoplus_{i=1}^m \mathcal{O}_{p_i}$ , and so the map is surjective and so  $h^1(C_m, \mathcal{O}_{C_m}(d)) = 0$ . Therefore, we have that  $h^0(C_m, \mathcal{O}_{C_m}) = md$ .  $\square$

We will need the following corollary in the next section.

cor:h00q

**Corollary 4.3.11.** *The global sections of the structure sheaf of a regular m-chain  $Q_m$  has dimension  $h^0(Q_m, \mathcal{O}_{Q_m}(d)) = md - 1$ , for  $d \geq 1$ .*

*Proof.* The proof is similar to that of Lemma 4.3.10, only now we have the short exact sequence

$$0 \rightarrow \mathcal{O}_{Q_m}(d) \rightarrow \bigoplus_{i=1}^m \mathcal{O}_{l_i}(d) \rightarrow \bigoplus_{i=1}^{m-1} \mathcal{O}_{p_i} \rightarrow 0,$$

since we only have  $m-1$  intersection points in an  $m$ -chain. Following the same argument as in the proof of Lemma 4.3.10, we get that  $h^0(Q_m, \mathcal{O}_{Q_m}(d)) = h^0(\bigoplus_{i=1}^m \mathcal{O}_{l_i}(d)) - h^0(\bigoplus_{i=1}^{m-1} \mathcal{O}_{p_i}) = md - 1$ .  $\square$

Sjekk om dette stemmer. Tror at strukturknippene ser annerledes ut når linjer bare snitter en annen.

prop:m-gon

**Proposition 4.3.12.** *Let  $C_m$  be a regular  $m$ -gon in  $\mathbb{P}^3$ . Then we can find a hypersurface  $S_d$  of degree  $d$  such that  $C_m \subseteq S_d$  when  $m < \binom{d+3}{3} \cdot \frac{1}{d}$ , or equivalently  $m < \frac{(d+3)(d+2)(d+1)}{6d}$ .*

*Proof.* By abuse of notation, we write  $\mathcal{O}_{C_m}$  for  $i_*\mathcal{O}_{C_m}$ , where  $i : C_m \rightarrow \mathbb{P}^3$  is the inclusion. We are interested in the global sections of twists of the ideal sheaf  $\mathcal{I}_{C_m}$  of the  $m$ -gon, i.e.  $H^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d))$ , to see if the  $m$ -gon can be contained in a hypersurface of degree  $d$ . To see why, consider the short exact sequence

$$0 \rightarrow \mathcal{I}_{C_m} \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_{C_m} \rightarrow 0$$

where  $\mathcal{I}_{C_m}$  is the kernel of the restriction map  $\mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_{C_m}$ . Since  $\mathcal{O}_{\mathbb{P}^3}(d)$  is locally free, we can twist with these to get the short exact sequence

$$0 \rightarrow \mathcal{I}_{C_m}(d) \rightarrow \mathcal{O}_{\mathbb{P}^3}(d) \rightarrow \mathcal{O}_{C_m}(d) \rightarrow 0.$$

From this we get the long exact sequence in cohomology:

$$\begin{array}{ccccccc} 0 & \rightarrow & H^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) & \rightarrow & H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) & \rightarrow & H^0(C_m, \mathcal{O}_{C_m}(d)) \\ & & \searrow & & \searrow & & \searrow \\ & & H^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) & \rightarrow & H^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) & \rightarrow & H^1(C_m, \mathcal{O}_{C_m}(d)) \\ & & \searrow & & \searrow & & \searrow \\ & & H^2(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) & \rightarrow & H^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) & \longrightarrow & \dots \end{array}$$

By Theorem 18.23 in <sup>E025</sup>[EO25], we know that  $H^i(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) = 0$  for  $i > 0$  and  $d \geq (-3)$ . Hence, we can focus on the first part of the sequence:

$$0 \rightarrow H^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \rightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \rightarrow H^0(C_m, \mathcal{O}_{C_m}(d)) \rightarrow H^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \rightarrow 0.$$

Now,  $H^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d))$  is precisely the kernel of the restriction map  $H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \rightarrow H^0(C_m, \mathcal{O}_{C_m}(d))$ , and so defines the global sections of  $\mathcal{O}_{\mathbb{P}^3}(d)$  which vanish on  $C_m$ . Therefore, the dimension of this space reveals whether the space of global sections of  $\mathcal{O}_{\mathbb{P}^3}(d)$  which vanish on  $C_m$  is non-empty or not. Taking dimensions in cohomology is additive, and we get that

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) + h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) - h^0(C_m, \mathcal{O}_{C_m}(d)).$$

By Theorem 18.4 (!?) in <sup>E025</sup>[EO25] we know  $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) = \binom{d+3}{3}$ . By Lemma 4.3.10 we have that  $h^0(C_m, \mathcal{O}_{C_m}(d)) = md$ . So we have that

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = \binom{d+3}{3} - md + h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)).$$

$H^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \geq 0$  measures the failure of the restriction map  $H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \rightarrow H^0(C_m, \mathcal{O}_{C_m}(d))$  to be surjective, i.e. the failure of lifting sections on  $C_m$  to global sections. If the map was surjective, we would have that  $h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = \binom{d+3}{3} - md$ . We can at least say that  $h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \geq 0$ , and so

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \geq \binom{d+3}{3} - md.$$

Meaning we can guarantee a degree  $d$  surface when  $\binom{d+3}{3} - md > 0$ , or equivalently when  $m < \frac{(d+3)(d+2)(d+1)}{6d}$ . □

*Remark 4.3.13.* Note that this argument does not guarantee that the surface is smooth, only that it exists.

To get an exact value for  $h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d))$ , we would need to compute  $h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d))$ . We have no direct answer of what this cohomology is, but the issue will be explored further through examples in the next two sections. We add this corollary to the proposition:

cor:h0-h1

**Corollary 4.3.14.**

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) - h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = \binom{d+3}{3} - md.$$

### 4.3.1 Computational results

Discuss numInIz, and also rename.

We have done computational experiments in Macaulay2 to get a better understanding of the inequality in Proposition 4.3.12. The script tests the existence and smoothness of hypersurfaces including an  $m$ -gon for various degrees  $d$  and number of lines  $m$ . The code used for these experiments is included in Section A.1, where we also discuss the details of the code. We note that to run this in Macaulay2, we have run the computation over a finite field  $\mathbb{Z}_{32749}$ , and over  $\mathbb{Q}$ , the latter being significantly slower.

We summarize the steps taken in the code for checking if an  $m$ -gon is included in some degree  $d$  surface:

1. Choose field  $\mathbb{C}$ , we use either  $\mathbb{Q}$  or  $\mathbb{Z}_{32749}$ , where the latter is faster, but not of characteristic 0.
2. Pick  $m$  random points in  $\mathbb{P}^3$ .
3. Build an  $m$ -gon by connecting the points in a cycle.
4. Connect pairs of points to generate the ideal of the line defined by them.
5. Intersect the ideals of the lines to get the ideal of the  $m$ -gon.
6. Pick a random element of degree  $d$  in the ideal of the  $m$ -gon.
7. Check if this element is smooth, singular, or equal to 0.

The results are listed in full in the appendix. The tables are included in Table A.1 and Table A.2, and we have included a smaller table of results over  $\mathbb{Q}$  in Table 4.1. They are consistent with the inequality in Proposition 4.3.12, always finding that an  $m$ -gon can be included in a surface of degree  $d$  when  $m$  is lower than the bound, and also always find the surfaces to be smooth. Interestingly, they also imply that we *cannot* find any surface (smooth or singular) containing an  $m$ -gon when  $m$  equals or exceeds the bound. This suggests that  $h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d))$  does not exceed  $md - \binom{d+3}{3}$  when  $m$  gets bigger, though we have not been able to prove this. Our tables include two values of  $m$  for each degree  $d$ . This is to check both below and above the bound for each degree. If the bound is an integer  $M$ , we test  $m = M - 1$ , and  $m = M$ . As is shown in the table in the appendix, we have found no cases where we can find a surface including an  $m$ -gon when  $m$  is equal to or larger than the bound.

**Table 4.1:** Results from Macaulay2 over  $\mathbb{Q}$  testing existence and smoothness of surfaces containing an  $m$ -gon. Each row corresponds to one experiment: degree  $d$ , number of points used, whether the chosen form was singular, whether the space of degree- $d$  forms in the ideal was zero, and the number of elements found in  $I_Z(d)$ .

mgon-results-small

| $d$ | $m$ | Singular | Zero  | numInIz |
|-----|-----|----------|-------|---------|
| 2   | 4   | false    | false | 2       |
| 2   | 5   | false    | true  | 0       |
| 3   | 6   | false    | false | 2       |
| 3   | 7   | false    | true  | 0       |
| 4   | 8   | false    | false | 3       |
| 4   | 9   | false    | true  | 0       |
| 5   | 11  | false    | false | 1       |
| 5   | 12  | false    | true  | 0       |
| 6   | 13  | false    | false | 6       |
| 6   | 14  | false    | true  | 0       |

*Remark 4.3.15.* When running the computations in Macaulay2, we have never encountered a case where we have gotten a singular surface one time, and a smooth surface another, even though the surfaces are picked at random. One of the reasons this is perhaps not very surprising, is to consider that being smooth is a Zariski open requirement, meaning the subset of smooth surfaces is open in the set of surfaces. Since we know that this means the subset is dense in the set of surfaces, and so it is not very surprising that the random surface picked is smooth. Still, this adds another reason to not use Macaulay2 as a proof of something, but merely a suggestion that it might be true. One could technically be very unlucky and only pick singular surfaces.

Talk about all the limitations of using the computational approach. Like: We find no surfaces above the bound, find only smooth surfaces, etc. When is this dependent on the field?

### 4.3.2 Cohomology of a 4-gon

c:4-gon-cohomology

As mentioned earlier, to get an exact value for  $h^0(\mathbb{P}^3, \mathcal{I}_{C_m})$ , we want to compute  $h^1(\mathbb{P}^3, \mathcal{I}_{C_m})$ . We will now explore this issue for the case of a 4-gon and looking at a suitable resolution of the ideal sheaf of the 4-gon. First we claim that the intersection points of a 4-gon can, by an automorphism of  $\mathbb{P}^3$ , be expressed as the points  $(1 : 0 : 0 : 0), (0 : 1 : 0 : 0), (0 : 0 : 1 : 0), (0 : 0 : 0 : 1)$ . Then the 4-gon can be expressed as the union of the lines  $l_1 = V(x_2, x_3), l_2 = V(x_0, x_3), l_3 = V(x_0, x_1), l_4 = V(x_1, x_2)$ . Now the ideal of the 4-gon is given by  $I_{C_4} = (x_2, x_3) \cap (x_0, x_3) \cap (x_0, x_1) \cap (x_1, x_2) = (x_1 x_3, x_0 x_2)$ . We justify our choice of points in the following lemma.

lem:gen-points

**Lemma 4.3.16.** *Let  $C_4$  be a regular 4-gon in  $\mathbb{P}^3$ . Then there exists an automorphism of  $\mathbb{P}^3$  such that the intersection points of  $C_4$  are given by the points  $(1 : 0 : 0 : 0), (0 : 1 : 0 : 0), (0 : 0 : 1 : 0), (0 : 0 : 0 : 1)$ .*

*Proof.* Let  $p_1, p_2, p_3, p_4$  be the intersection points of  $C_4$ . To be in general position in  $\mathbb{P}^3$ , we require that no three of the points are collinear, and that the points do not all lie in a plane (The second part is already fulfilled by considering a regular 4-gon, since if this was included in a plane, then all lines would intersect!). Let  $A$  be the  $4 \times 4$  matrix whose columns are the homogeneous coordinates of the points. Since we have that the points do not all lie on a plane, we know that the rows are linearly independent, and we



can transform the points by multiplying the matrix by  $A^{-1}$ . Then  $AA^{-1} = I_4$ , and we have sent the points to the desired locations.  $\square$

To get a value of  $h^1(\mathbb{P}^3, \mathcal{I}_{C_4})$ , we will first look at a resolution of the ideal sheaf  $I_{C_4}$ . We take inspiration from Hilbert's syzygy theorem, [EO25] Theorem 17.16, which states that we can find a finite free resolution for a finitely generated graded module. We therefore aim to find a resolution. From the discussion above, we have that  $I_{C_4} = (x_1x_3, x_0x_2)$  (the calculation was done in Macaulay2). We know that  $I_{C_4}$  is of the form  $\tilde{M}$  for some graded  $R$ -module  $M$ , with  $R = \mathbb{C}[x_0, x_1, x_2, x_3]$ , see [EO25], theorem 17.9. We can therefore look at a free resolution of  $M$  to get a resolution of  $I_{C_4}$ . First, consider the map

$$R^{\oplus 2}(-2) \rightarrow M$$

given by the matrix

$$\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}.$$

This map is surjective, as we see that the image contains the generators of  $M$ . Now, consider  $(f, g) \in R^{\oplus 2}(-2)$ , which maps to  $x_0x_2f + x_1x_3g$ . Then we know that at least  $f = x_1x_3$  and  $g = -x_0x_2$  maps to zero, and so we continue the resolution:

$$R(-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} R(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} M \rightarrow 0.$$

Now we want to check if these are the only elements of the kernel. If  $x_0x_2f + x_1x_3g = 0$ , we have that  $x_0x_2f = -x_1x_3g$ . Since  $x_0, x_1, x_2, x_3$  are all non-zero divisors in  $R$ , we must have that  $f$  is divisible by  $x_1x_3$  and  $g$  is divisible by  $x_0x_2$ . Therefore, we can write  $f = x_1x_3h$  and  $g = -x_0x_2h$  for some  $h \in R$ . This means that the kernel of the map is generated only by the element we found, and so the resolution is exact. Then we have the following free resolution of  $M$ :

$$0 \rightarrow R(-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} R(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} M \rightarrow 0.$$

Passing to the associated sheaves, we get the following resolution of the ideal sheaf of the 4-gon:

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^3}(-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4} \rightarrow 0.$$

Now we can twist this resolution by  $\mathcal{O}_{\mathbb{P}^3}(d)$  to get

$$0 \rightarrow \mathcal{O}_{\mathbb{P}^3}(d-4) \xrightarrow{\begin{pmatrix} x_1x_3 \\ -x_0x_2 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(d-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4}(d) \rightarrow 0. \quad (4.2) \quad \boxed{\text{eq:resol}}$$

From this, we can get the long exact sequence in cohomology:

$$\begin{array}{ccccccc} 0 & \rightarrow & H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d-4)) & \rightarrow & H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d-2)^{\oplus 2}) & \rightarrow & H^0(\mathbb{P}^3, I_{C_4}(d)) \\ & & \searrow & & \searrow & & \searrow \\ & & H^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d-4)) & \rightarrow & H^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d-2)^{\oplus 2}) & \rightarrow & H^1(\mathbb{P}^3, I_{C_4}(d)) \\ & & \searrow & & \searrow & & \searrow \\ & & H^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d-4)) & \rightarrow & \dots & & \end{array}$$

By Theorem 18.23 in [E025], we know that  $H^i(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) = 0$  for  $i > 0$  and  $d \geq (-3)$ . Therefore, when  $d - 4 \geq -3$ , i.e.  $d \geq 1$ , we have that  $H^1(\mathbb{P}^3, I_{C_4}(d)) = 0$ . Thus, we get the following proposition:

**Proposition 4.3.17.** *Let  $C_4$  be a regular 4-gon in  $\mathbb{P}^3$ . Then for  $d \geq 1$ , we have that  $h^1(\mathbb{P}^3, \mathcal{I}_{C_4}(d)) = 0$ , and we can find a degree  $d$  surface containing  $C_4$  for  $d \geq 2$ .*

*Proof.* We have shown above that for  $d \geq 1$ ,  $H^1(\mathbb{P}^3, I_{C_4}(d)) = 0$ . Then the second part follows directly from Corollary 4.3.14 by the following observation: For  $d = 1$ , we have that  $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) = \binom{4}{3} - 4 = 0$ , so there are no degree 1 surfaces (planes) containing a general 4-gon.  $\square$

This is perhaps unsurprising, and in fact we could already show that a regular 4-gon can be included in a degree 2 surface, for example a smooth quadric surface:

cor:4-gon

**Corollary 4.3.18.** *Let  $C_4$  be a regular 4-gon in  $\mathbb{P}^3$ . Then we can find a smooth degree 2 surface  $S_2$  such that  $C_4 \subseteq S_2$ .*

*Proof.* Using the points from Lemma 4.3.16, we can include the 4-gon in  $V(x_1x_3 - x_0x_2)$ , and by generality of these points we can find such a smooth quadric surface for any general 4-gon.  $\square$

Still, there are interesting conclusions to be drawn from this. For example, we can find  $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d))$  for all  $d \geq 1$ , and therefore know the exact dimension of the space of degree  $d$  surfaces containing a regular 4-gon:

**Corollary 4.3.19.** *Let  $C_4$  be a regular 4-gon in  $\mathbb{P}^3$ . Then for  $d \geq 2$ , we have that*

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_4}(d)) = \binom{d+3}{3} - 4d.$$

For example, for  $d = 2$ , we have that  $h^0(\mathbb{P}^3, \mathcal{I}_{C_4}(2)) = \binom{5}{3} - 8 = 10 - 8 = 2$ , so the linear system of quadrics containing a regular 4-gon is 1-dimensional(!?).

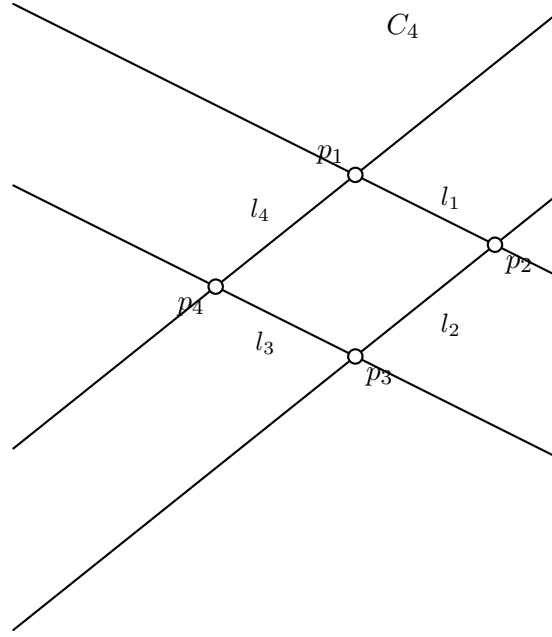
This section also reveals why we have not yet found a general result for  $h^1(\mathbb{P}^3, \mathcal{I}_{C_m})$ , which is that we have so far relied on the generators of the ideal  $I_{C_4}$  to compute it. The generators of ideals are generally troublesome, and it is hard to prove general facts by using them. A concrete reason for why we had trouble generalizing in our problem, was that when checking the ideal sheaves of higher  $m$  in Macaulay2, we found that they do not grow linearly. We give a short table of the number of generators of the ideals of  $m$ -gons for small  $m$ :

Table 4.2: Number of generators of the ideals of  $m$ -gons for small  $m$ .

tab:num-generators

| $m$ | Number of generators |
|-----|----------------------|
| 4   | 2                    |
| 5   | 5                    |
| 6   | 5                    |
| 7   | 7                    |
| 8   | 7                    |
| 9   | 11                   |
| 10  | 6                    |
| 11  | 15                   |

Note that these are the (minimal) number of all generators, not just for degree  $d$ .

Figure 4.3: A 4-gon  $C_4$  in  $\mathbb{P}^3$ , drawn on a plane.

## 4.4 Surfaces including m-chains

A ‘happy accident’ we made when studying  $m$ -gons in Macaulay2 was to forget to add the final line which then intersects the first line, closing the loop. Before realizing what exactly had happened, we noted that the results were strange and suspected an error in our code, which we then promptly fixed. Still, the results were a bit surprising, implying that for some chains we find only singular surfaces of certain degrees. After getting more expected results for  $m$ -gons, we decided to study these new formations further, resulting in the theory in this section. We begin by defining  $m$ -chains.

**Definition 4.4.1.** An  $m$ -chain is a union of  $m$  lines  $l_1, l_2, \dots, l_m$  such that  $l_i$  intersects  $l_{i+1}$  for  $1 \leq i < m$ . We will write  $Q_m = \bigcup_{i=1}^m l_i$  and denote the intersection points as  $p_i = l_i \cap l_{i+1}$  for  $1 \leq i < m$ .

**Definition 4.4.2.** A *regular*  $m$ -chain is an  $m$ -chain such that no line  $l_i$  intersects any  $l_j$  for  $|i - j| > 1$ , and there are no double lines  $l_i \neq l_j$  for  $i \neq j$ .

**Definition 4.4.3.** We define  $Q_m \subseteq \mathbb{G}(1, 3)^m$  to be the subset of  $m$ -tuples of lines  $(l_1, \dots, l_m)$  such that they form an  $m$ -chain.

*Remark 4.4.4.* We showed in Lemma 4.3.6 that  $Q_m$  is an irreducible variety of dimension  $3m + 1$  for  $m \geq 1$ .

**Corollary 4.4.5.** The set of regular  $m$ -chains in  $Q_m$  is a dense open subset of  $Q_m$ .

*Proof.* The proof is the same as that for Lemma 4.3.9. We are able to replace  $m$ -gons with  $m$ -chains since the set of  $m$ -chains is the same as  $m$ -gons with the non-destruction exclusion that  $l_m$  need not intersect  $l_1$ .  $\square$

As discussed above, we note that an  $m$ -chain is an  $(m + 1)$ -gon excluding the line  $l_m$  intersecting  $l_1$ . One can then modify the code to create  $m$ -chains instead of  $(m + 1)$ -gons, which is included in Section A.2. The results from this are similar to those of  $m$ -gons,

in that for higher degrees, we find smooth surfaces, up until we find no surfaces. There are however interesting exceptions for degree 2 and 3 surfaces. Here we find singular surfaces when considering 4-chains and 6-chains respectively. We will handle the cases separately. First, we show a similar proposition to that of  $m$ -gons in Proposition 4.3.12.

prop:n-chain

**Proposition 4.4.6.** *Let  $Q_m$  be a regular  $m$ -chain in  $\mathbb{P}^3$ . Then we can find a hypersurface  $S_d$  of degree  $d$  such that  $Q_m \subseteq S_d$  when  $m < ((\binom{d+3}{3} - 1) \cdot \frac{1}{d})$ , or equivalently  $m < \frac{(d+3)(d+2)(d+1)-1}{6d}$ .*

*Proof.* The proof is very similar to that of Proposition 4.3.12, so we will be brief here. Again, by abuse of notation, we write  $\mathcal{O}_{Q_m}$  for  $i_*\mathcal{O}_{Q_m}$ , where  $i : Q_m \rightarrow \mathbb{P}^3$  is the inclusion. We have the short exact sequence

$$0 \rightarrow \mathcal{I}_{Q_m}(d) \rightarrow \mathcal{O}_{\mathbb{P}^3}(d) \rightarrow \mathcal{O}_{Q_m}(d) \rightarrow 0,$$

which by the same analysis as before gives us

$$h^0(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) + h^1(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) - h^0(Q_m, \mathcal{O}_{Q_m}(d)).$$

From Corollary 4.3.11 we have that  $h^0(Q_m, \mathcal{O}_{Q_m}(d)) = md - 1$ . Again, by Theorem 18.4 (!?) in [EO25] we know  $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) = \binom{d+3}{3}$ . Thus, we have

$$h^0(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) = \binom{d+3}{3} - md - 1 + h^1(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)).$$

We still know that  $h^1(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) \geq 0$ , so

$$h^0(\mathbb{P}^3, \mathcal{I}_{Q_m}(d)) \geq \binom{d+3}{3} - md - 1,$$

and we can guarantee a surface when  $m < ((\binom{d+3}{3} - 1) \cdot \frac{1}{d})$ . □

#### 4.4.1 Computational results

Include m2 results for  $m$ -chains, including the unique surface including the 6-chain.

#### 4.4.2 The case of 4-chains

**Proposition 4.4.7.** *For a regular 4-chain in  $\mathbb{P}^3$ , we cannot find a smooth quadric surface  $S_2 \subset \mathbb{P}^3$  such that it includes the 4-chain.*

*Proof.* First, assume that  $S_2$  is smooth. Since  $S_2$  is a smooth quadric surface in  $\mathbb{P}^3$ , it is of the form  $S_2 = \mathbb{P}^1 \times \mathbb{P}^1$  (reference Beauville!). This means that  $S_2$  has two rulings, meaning that there are two families of lines on  $S_2$ , such that any two lines in the same family do not intersect, and any two lines in different families intersect at exactly one point. Denote the two families as  $F_1$  and  $F_2$ . Then we can see that a 4-chain must consist of two lines from  $F_1$  and two lines from  $F_2$ , since any two lines in the same family do not intersect. This means that the 4-chain must be of the form  $l_1, l_2, l_3, l_4$  where  $l_1, l_3 \in F_1$  and  $l_2, l_4 \in F_2$ . However, this means that  $l_4$  intersects  $l_1$ , since they are in different families, meaning that the 4-chain is in fact a 4-gon. This is a contradiction, since this is not a regular 4-chain. Thus, we cannot find a smooth quadric surface  $S_2 \subset \mathbb{P}^3$  such that it includes the 4-chain. □

This contrasts the case of 4-gons, where we could find smooth quadric surfaces including the 4-gon, as shown in Corollary 4.3.18 in Section 4.3.2.

lem:2planes

**Proposition 4.4.8.** *For a 4-chain in  $\mathbb{P}^3$ , we can find a singular quadric surface  $S_2 \subset \mathbb{P}^3$  such that it includes the 4-chain.*

*Proof.* We pair the lines of the 4-chain as  $(l_1, l_2)$  and  $(l_3, l_4)$ . First we assume that the 4-chain is regular. Each pair of lines defines a plane in  $\mathbb{P}^3$ , denoted as  $H_1 = \text{span}(l_1, l_2)$  and  $H_2 = \text{span}(l_3, l_4)$ . Since the 4-chain is regular, we have that  $H_1 \neq H_2$ . Then the union of the two planes  $S_2 = H_1 \cup H_2$  is a singular quadric surface in  $\mathbb{P}^3$  that includes the 4-chain.

Now we consider a non-regular 4-chain. If the lines are not distinct, we choose  $H_1$  and  $H_2$  to be any planes including the pairs of lines respectively. If the lines intersect more than the adjacent lines, the plane might be the same, in that case we choose  $H_2$  to be any plane different from  $H_1$ .  $\square$

In fact, this is the only quadric surface which includes a regular 4-chain, by an argument we will use in a similar result for 6-chains in cubic surfaces, specifically Proposition 4.4.16.

Sjekk om beviset utvides til 4-kjeder.

### 4.4.3 The case of 6-chains

Let  $\mathcal{M}$  be the set of smooth cubic surfaces in  $\mathbb{P}^3$ . This is an open subset of  $\mathbb{P}(\mathbb{C}[x_0, \dots, x_3]_3) \cong \mathbb{P}^{19}$ . Since  $\mathcal{M}$  is an open subset, it shares the dimension, i.e.  $\dim \mathcal{M} = 19$ . Let  $\mathcal{N} = \{(S_d, Q_6) \in \mathcal{M} \times \mathcal{Q}_6 \mid Q_6 \subset S_d\}$  be the incidence variety of smooth cubic surfaces including a regular 6-chain. Note that this is equivalent to the set  $\{(S_d, l_1, \dots, l_6) \in \mathcal{M} \times \mathbb{G}(1, 3)^6 \mid \bigcup_{i=1}^6 l_i \text{ is a regular 6-chain included in } S_d\}$ , it is just notationally simpler to use the first definition. We have the following diagram:

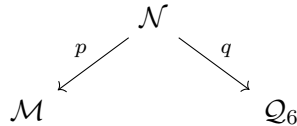


Figure 4.4: Incidence diagram with the morphisms  $p : \mathcal{N} \rightarrow \mathcal{M}$  and  $q : \mathcal{N} \rightarrow \mathcal{Q}_6$ .

g:incidence

*Remark 4.4.9.* Note that it is important that we consider the incidence variety over specifically the smooth cubic surfaces including a regular 6-chain. If not, one can prove that the fiber of the union of 3 planes, will include a much higher dimensional family of 6-chains, and so the incidence variety will not be irreducible. (Do calculation!?)

**Proposition 4.4.10.**  *$\mathcal{N}$  is an irreducible variety.*

Sjekk irreducibilitet.

*Proof.* We prove this inductively, by considering projection maps  $\pi_m : \mathcal{N}_m \rightarrow \mathcal{N}_{m-1}$  where  $\mathcal{N}_m$  is the incidence variety of smooth cubic surfaces including a regular  $m$ -chain. The base case is  $m = 1$ , where we have the incidence variety of smooth cubic surfaces including a line. This is irreducible by Proposition 2.6.4 in Section 2.6. Now, assume that  $\mathcal{N}_{m-1}$  is irreducible. Consider the projection map  $\pi_m : \mathcal{N}_m \rightarrow \mathcal{N}_{m-1}$  given by  $\pi_m(S_d, Q_m) = (S_d, Q_{m-1})$ , where  $Q_{m-1}$  is the  $(m-1)$ -chain obtained by removing the

last line from  $Q_m$ . Then the fiber over a point  $(S_d, Q_{m-1}) \in \mathcal{N}_{m-1}$  is given by the set of lines  $l_m$  such that  $Q_m = Q_{m-1} \cup l_m$  is an  $m$ -chain included in  $S_d$ . Since  $Q_{m-1}$  is a chain, it has a unique last line  $l_{m-1}$ , and so  $l_m$  must intersect  $l_{m-1}$ . Thus, the fiber is given by the set of lines in  $S_d$  intersecting  $l_{m-1}$ . This is an irreducible variety by modifying the proof of Lemma 4.3.5 to consider the Fano variety of lines in  $S_d$ ,  $F_1(S_d)$ , instead of the full Grassmannian  $\mathbb{G}(1, 3)$  (is this correct!?). Therefore, the fiber is also irreducible. Then by Lemma 1.3.4 we have that  $\mathcal{N}_m$  is irreducible.  $\square$

Now we will examine 6-chains in cubic surfaces. We will show that a regular 6-chain is not included in a smooth cubic surface in two ways(!?), but that we can find singular cubic surfaces including a regular 6-chain. We start by studying the incidence variety of smooth cubic surfaces including a regular 6-chain. First we show that we can find a singular cubic surface including a regular 6-chain, which will be useful to prove that we cannot find a smooth cubic surface including a regular 6-chain.

lem:3planes

**Lemma 4.4.11.** *Any 6-chain is included in a union of 3 planes in  $\mathbb{P}^3$ .*

*Proof.* As in Proposition 4.4.8 We pair the lines of the 6-chain as  $(l_1, l_2)$ ,  $(l_3, l_4)$  and  $(l_5, l_6)$ . Each pair of lines define a plane in  $\mathbb{P}^3$ . Then the union of the three planes is a singular cubic surface in  $\mathbb{P}^3$  that includes the 6-chain.  $\square$

Another lemma we will need is to show that we can find a 6-chain in any smooth cubic surface. We show this by finding a 12 chain. This will be quite degenerate, but we are not looking for a regular chain here.

lem:12

**Lemma 4.4.12.** *For a smooth cubic surface in  $\mathbb{P}^3$ , we can find a 12-chain such that it is included in the surface.*

*Proof.* We will build this 12-chain explicitly. Recall from Proposition 3.4.2 that a smooth cubic surface is isomorphic to the blowup of  $\mathbb{P}^2$  at six points, and that there are 27 lines on the surface, as described in Proposition 3.4.3. We will use the notation from these propositions. Consider the exceptional curves  $E_1, E_2, E_3, E_4, E_5, E_6$ , and the strict transforms of the lines  $\ell_{12}, \ell_{23}, \ell_{34}, \ell_{45}, \ell_{56}$  through pairs of points. Then one forms a 12-chain by considering the lines in the following order:

$$E_1, \ell_{12}, E_2, \ell_{23}, E_3, \ell_{34}, E_4, \ell_{45}, E_5, \ell_{56}, E_6, C_1, \ell_{16}.$$

We include a figure of this in Fig. 4.5 for clarity.  $\square$

Note that one could continue building the chain further, but we stop at 12 for simplicity, and especially for the nicer picture in Fig. 4.5. Note also that the 12-chain is also a 12-gon, since the last line intersects the first line.

prop:6chain

**Proposition 4.4.13.** *For a general 6-chain in  $\mathbb{P}^3$ , we cannot find a smooth cubic surface  $S_3 \subset \mathbb{P}^3$  such that it includes the 6-chain.*

*Proof.* Let  $p: \mathcal{N} \rightarrow \mathcal{M}$  be given by  $p(S_d, Q_6) = S_d$ . Then the fiber over a point  $[S_d] \in \mathcal{M}$  is given by the set of 6-chains in  $S_d$ . Since  $S_d$  is a smooth cubic surface, it includes exactly 27 lines, so we can find a 6-chain by choosing 6 lines from these 27. We know from Lemma 4.4.12 that it is possible to find a 6-chain, so this fiber is not empty. Furthermore, there are only a finite number of lines, and so the fiber  $p^{-1}(S_d)$  has dimension 0. Thus, from the dimension formula for fibers, Theorem 1.3.5, we have that  $\dim \mathcal{N} = \dim \mathcal{M} + \dim p^{-1}(S_d) = 19$ .

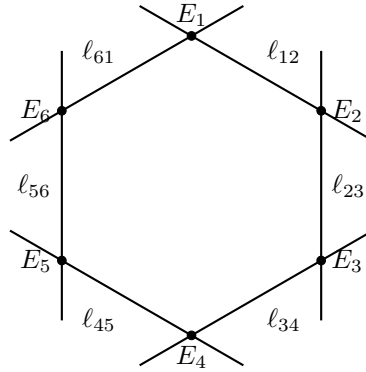


Figure 4.5: Six exceptional lines  $E_1, \dots, E_6$ , and the strict transform of six lines, used to form a 12-gon.

fig:12chain

Now we are interested in  $q$ , and whether this dominates  $\mathcal{Q}_6$ . If it does not, we know that a general 6-chain does not have a representative in  $\mathcal{N}$ , and so is not included in a smooth cubic surface. By Lemma 4.3.6, we have that  $\dim \mathcal{Q}_6 = 19$ . Since the dimension of these sets are the same, we aim to show that the fiber is not 0-dimensional, so that the dimension formula would imply that the dimension of  $\mathcal{N} > 19$  if  $q$  dominates  $\mathcal{Q}_6$ . Hence, we claim that if a 6-chain is included in a smooth cubic surface, it must necessarily be included in at least a 1-dimensional family of cubic surfaces. To see this, let  $Q_6$  be a general 6-chain in  $\mathbb{P}^3$  which is included in a smooth surface  $X$ . Then, from Lemma 4.4.11, we know that it is also included in a singular surface  $Y$ , since we made no restrictions on the 6-chain in the lemma. Then let  $\mathcal{D}$  be the pencil  $\{aX + bY\}$ . All the surfaces in this linear system include our 6-chain, by the observation that if  $X = V(f)$  and  $Y = V(g)$ , then inclusion is equivalent to  $f|_{Q_6} = g|_{Q_6} = 0$ , and  $af|_{Q_6} + bg|_{Q_6} = 0$ . Then we apply Bertini (that smoothness is an open property!?) to see that most of the surfaces are smooth. This gives us a 1-dimensional family of smooth surfaces including our 6-chain, and so  $q$  cannot dominate  $\mathcal{Q}_6$ .  $\square$

*Remark 4.4.14.* We use the set of  $m$ -chains, not regular  $m$ -chains, so that the fiber of  $p$  is non-empty.

Note that this means that the 6-chains which *are* included in smooth cubic surfaces, are not general. To see why this might be, note that as we saw in the proof of Lemma 4.3.9, requiring that pairs of lines in  $\mathbb{P}^3$  do not meet is a Zariski-open condition. This is exactly the separating factor of the 6-chains included in smooth cubic surfaces. The requirement of non-intersection is also included in the definition of a regular  $m$ -chain.

The fact that we cannot find a surface including a general 6-chain may not be very intuitive, but the following lemma gives a good reason.

**Lemma 4.4.15.** *For a smooth cubic surface, we can find at most a regular 5-chain.*

*Proof.* First we show that we can find a regular 5-chain. Again we consult Fig. 3.1 which shows a smooth surface as the blow-up  $\mathbb{P}^2$  in six points. Then the chain  $E_1 \cup \ell_{12} \cup E_2 \cup \ell_{23} \cup E_3$  meets the criteria of a regular 5-chain.

Now we want to show that we cannot find a regular 6-chain. First, denote the 27 lines of the cubic surface as  $C_1, \dots, C_{27}$ . Let  $Q_5$  be a regular 5-chain, consisting of  $C_1, \dots, C_5$ . Look at the class  $D = C_1 + C_2 - C_4 - C_5$ . Then  $D^2 = 0$ , and  $HD = 0$  for  $H = f^*(\mathcal{O}_{\mathbb{P}^3}(1))(!?)$ . Since  $H$  is ample, the Hodge index theorem, Theorem 3.1.8, tells

us that  $D = 0$ , and we have the equality  $C_1 + C_2 = C_4 + C_5$ . Let  $C_i \neq C_4$  be another line which intersects  $C_5$ . Then it must also intersect either  $C_1$  or  $C_2$  by the equality, and thus cannot be used to make a regular 6-chain.  $\square$

We have only used the assumption that the  $m$ -chain do not intersect extra, so rewrite to clarify that this holds for more than just the regular chains.

Similar result for gons?

One could also use this lemma to prove Proposition 4.4.13 for regular 6-chains, similarly to our proof that a regular 4-chain is not included in a smooth quadric surface. i.e. we note that if we have a 6-chain in a smooth cubic surface, the lemma above tells us that this cannot be a regular 6-chain.

Our testing in Macaulay2 also suggested that the singular surface was unique, which we will prove in the following proposition.

prop:unique

**Proposition 4.4.16.** *A regular 6-chain in  $\mathbb{P}^3$  is only included in a unique surface.*

Sjekk  $q$  projektiv.

*Proof.* Consider the diagram in Fig. 4.4. The map  $q$  has fiber  $q^{-1}(Q_6) = \{S \mid Q_6 \in S\}$  for  $Q_6 \in \mathcal{Q}_6$ .  $q$  is a projective morphism, since it is the composition of the closed embedding  $i: \mathcal{N} \rightarrow \mathbb{P}^N \times \mathcal{M}$ , and the projective map  $\pi: \mathbb{P}^N \times \mathcal{M} \rightarrow \mathcal{M}$ . A projective morphism is proper by [Har77] Theorem II.4.9. Then taking the dimension of the fiber of  $q$  is upper semi-continuous.

From (refer to result in Macaulay2 saying that the surface is unique!), we have an example where the fiber is a single point. Call the relevant 6-chain  $Z$ . Though this computation was done over  $\mathbb{Q}$ , the result extends to  $\mathbb{C}$ . To see why, first note that when we find that there is only one unique surface including  $Z$ , it is equivalent to saying that the kernel of the map  $\phi: H^0(\mathbb{P}^3, \mathcal{O}(3)) \rightarrow H^0(Z, \mathcal{O}(3)|_Z)$  is 1-dimensional (0-dimensional!). Note also that the fiber of  $q(Z)$  is the set  $\{f \in \mathbb{C}[x_0, x_1, x_2, x_3]_3 \mid f|_Z = 0\}$ , which are the polynomials which are zero on  $Z$ . The equations requiring  $f|_Z = 0$  have rational coefficients, since our  $Z$  is defined over the rationals. Therefore, the kernel of  $\phi$  is also 1-dimensional over  $\mathbb{C}$ . Because of this, there exists at least one 6-chain with a single point as fiber, and so the fiber over this point is 0-dimensional.

By Lemma 4.4.11, we know that any 6-chain is included in at least one cubic surface, so the fiber of  $q$  is non-empty for all points in  $\mathcal{Q}_6$ . Then, by upper semi-continuity of fiber dimension (add to basic results, and cite!), we have that since we know one fiber,  $q^{-1}(Z)$  is 0-dimensional, there exists an open neighborhood  $U$  of  $Z$ , such that  $\dim q^{-1}(Z') \leq 0$  for all  $Z' \in U$ . Since every fiber is non-empty, all of these fibers are 0-dimensional, and so there are a finite number of surfaces including a regular  $Q_6$ . Then we note that the fiber of  $q$  is the set of solutions of linear equations, and so it defines a projective space. Since the fiber is 0-dimensional, it must be the point  $\mathbb{P}^0$ , and so there is only one surface.  $\square$



## Chapter 5

# Conclusion

conc

Vurder å skrive conclusion hvor jeg diskuterer resultater og legger til formodninger og videre arbeid.

**Conjecture 5.0.1.** *Let  $C_m$  be a regular  $m$ -gon in  $\mathbb{P}^3$ , then we have*

$$h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = \max \left( 0, \binom{d+3}{3} - md \right).$$

This would mean that the bound we have found for the existence of surfaces containing the  $m$ -gon is sharp, and we would get the corollary:

**Corollary 5.0.2.** *For a regular  $m$ -gon  $C_m$  in  $\mathbb{P}^3$ , we can not find any degree  $d$  surface  $S_d$  such that  $C_m \subseteq S_d$  when  $m \geq \binom{d+3}{3} \cdot \frac{1}{d}$ .*

### 5.1 Sjekk til slutt

- Legg til at vi jobber med "regulære" ting alle plasser vi gjør det.
- Bestem mellom  $\mathbb{C}$  og  $\mathbb{k}$ ?
- Gå gjennom alle plasser jeg har skrevet 'amount' og bytt til 'number'.
- Gå over  $Q_m$   $Z_m$  osv. og sørg for at det er konsekvent.
- Gjør referanser til flater konsekvent. Mye bytting mellom  $S$ ,  $S_3$ ,  $X$  tror jeg.
- Endre fra  $\pi_1, \pi_2$  til  $p, q$  for projeksjoner fra Incidence.
- Fjern tid fra tabell.
- Endre inline til equation?
- Legg til  $i_*$  (pushforward).
- Endre 'Schubert variety  $\Sigma_L$ ', da det bør være  $\Sigma_1$ . Dette er det samme som en Schubert varietet, men det er ikke det vi definerer det som.
- Sjekk at vi skriver smooth variety i intersection theory.
- Oppdater alle referanser til EO.
- Gå over "".
- [reference]



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## References

# Appendix A

## Computational methods and results

appendix

This appendix documents the Macaulay2 scripts used for the computational experiments. We describe how the line configurations are built, what the scripts test, and how to interpret the tables. We also note limitations of the approach (randomness, characteristic, layout).

### A.1 The $m$ -gon script

app:mgon

An  $m$ -gon is produced by choosing  $m$  random points of  $\mathbb{P}^3$  and declaring lines through consecutive pairs, plus the final pair  $(p_m, p_1)$ . The points thus model the pairwise intersections required in the cycle.

lst:m-gon

Listing A.1: Script generating an  $m$ -gon and testing degree- $d$  forms

```
1 -- Step 1: Define the ring and parameters
2 kk = QQ;
3 RingP3 = kk[x_0..x_3];
4 d = 2; -- Degree of the hypersurface
5 nrPoints = 12;
6
7 -- Step 2 and 3: Generate random points and create pairs of
  subsequent points
8 -- Function for generating i random points in P^3
9 randomPoints = i -> (
10   v := for j in 0..3 list random(kk);
11   -- Ensure the point is not zero
12   while all(v, x -> x == 0) do (
13     v = for j in 0..3 list random(kk)
14   );
15   v
16 )
17 -- Pick nrPoints random points in P^3
18 myPoints = apply(nrPoints, randomPoints);
19 -- Create pairs of subsequent points, and also the pair consisting
  of the last and first point
20 subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i
  , myPoints#(i+1)};
21 -- Add final pair
22 subsequentPairs = append(subsequentPairs, {myPoints#0, myPoints#(
  myPoints-1)});
23
```

## Appendix A. Computational methods and results

```

24 -- Step 3: Generate matrices from point pairs
25 -- Function to convert two points to a matrix with first row {x_0
    .. x_3}
26 pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points
    #1};
27 -- Convert each pair of points to a matrix
28 pointsMatrix = apply(subsequentPairs, pointsToMatrix);
29
30 -- Step 4: Get the ideals of the lines
31 -- Given a matrix with first row {x_0 .. x_3} and next two rows
    points, return the ideal of the line through the points
32 lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
33 allLines = apply(pointsMatrix, lineFromPoints);
34
35 -- Step 5: Create the ideal of the union of all lines
36 -- Intersect the ideals of the lines to get ideal of the union of
    all lines
37 Iz = intersect allLines;
38 IzGens = mingens Iz; -- If one wants to see the number of
    generators
39 -- hf = hilbertFunction(d, RingP3/Iz);
40 -- totalForms = binomial(d+3,3);
41 -- numInIz = totalForms - hf;
42
43 -- Step 6: Pick a degree d element in Iz, which defines a surface
    containing all the lines, and check properties
44 -- Pick a random degree d element in Iz
45 f = random(d, Iz);
46 -- Create the variety defined by f
47 Vf = variety ideal f;
48
49 -- Check if the surface is smooth and not the zero polynomial
50 codimension = codim singularLocus ideal f;
51 isZero = (f == 0)
52 isSingular = (codimension < 4)
53 isSmooth(Vf) and not isZero
54
55 IzGens
56 -- numInIz

```

Workflow (enumerated):

1. **Setup** Choose a field  $k$  ( $\mathbb{Q}$  or  $\mathbb{Z}_{32749}$ ) and work in  $R = k[x_0, x_1, x_2, x_3]$ .
2. **Random points** Pick  $m$  nonzero vectors in  $k^4$ ; interpret them as homogeneous points  $p_1, \dots, p_m$ .
3. **Pairs** Form  $(p_i, p_{i+1})$  for  $1 \leq i < m$  and add  $(p_m, p_1)$ .
4. **Matrix to line** Each ordered pair yields a  $3 \times 4$  matrix

$$((a, b, c, d), (e, f, g, h)) \mapsto \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ a & b & c & d \\ e & f & g & h \end{pmatrix}$$

whose  $3 \times 3$  minors generate the ideal of the line through the points.

5. **Union ideal** Intersect the  $m$  line ideals to get  $I_Z$  for  $Z = \bigcup l_i$ .
6. **Degree- $d$  test** Inspect  $I_Z(d)$ . If nonzero, pick a random element and test smoothness via the Jacobian criterion; if the element returned is 0 retry until nonzero (a safeguard easily added).

*Remark A.1.1* (Zero element safeguard). Random selection almost never returns 0 when  $I_Z(d) \neq 0$ , but to be rigorous one can loop until a nonzero form is obtained or a retry limit is hit.

The computational goal: determine largest  $m$  for which  $I_Z(d) \neq 0$  and a generic form is smooth, comparing with the theoretical bound in Proposition [4.3.12](#). <sup>prop:m-gon</sup>

### A.1.1 Tables of results

We show values just below and at/above the theoretical bound. "Zero" indicates  $I_Z(d) = 0$ . We used `\FloatBarrier` in the main document to encourage tables to appear together.

Table A.1: Results over  $\mathbb{Q}$  for  $m$ -gons.

| oprule | $d$ | $m$ | Singular | Zero  | numInIz |
|--------|-----|-----|----------|-------|---------|
|        | 2   | 4   | false    | false | 2       |
|        | 2   | 5   | false    | true  | 0       |
|        | 3   | 6   | false    | false | 2       |
|        | 3   | 7   | false    | true  | 0       |
|        | 4   | 8   | false    | false | 3       |
|        | 4   | 9   | false    | true  | 0       |
|        | 5   | 11  | false    | false | 1       |
|        | 5   | 12  | false    | true  | 0       |
|        | 6   | 13  | false    | false | 6       |
|        | 6   | 14  | false    | true  | 0       |

## A.2 The m-chain script

An  $m$ -chain omits the closing line: with points  $p_1, \dots, p_m$  we form lines only between  $p_i$  and  $p_{i+1}$  for  $1 \leq i < m$ . Thus there are  $m - 1$  lines. All other steps mirror the  $m$ -gon case; when comparing bounds replace  $m$  (lines) by  $m - 1$ .

Listing A.2: Script for an  $m$ -chain (number of lines =  $m - 1$ )

```

1 -- Step 1: Define the ring and parameters
2 -- kk = QQ;
3 kk = ZZ/32749;
4 RingP3 = kk[x_0..x_3];
5 d = 2; -- Degree of the hypersurface
6 nrLines = 5;
7 nrPoints = nrLines+1;
8 maxNoPoints = (binomial(d+3, 3) - 1)/d;
9
10 -- Step 2 and 3: Generate random points and create pairs of
    subsequent points
11 -- Function for generating i random points in P^3
12 randomPoints = i -> (
```

**Table A.2:** Results over  $\mathbb{Z}_{32749}$  for  $m$ -gons (same columns).

gon-results\_Z32749

| opr | rule | $d$ | $m$ | Singular | Zero  | numInIz |
|-----|------|-----|-----|----------|-------|---------|
|     |      | 2   | 4   | false    | false | 2       |
|     |      | 2   | 5   | false    | true  | 0       |
|     |      | 3   | 6   | false    | false | 2       |
|     |      | 3   | 7   | false    | true  | 0       |
|     |      | 4   | 8   | false    | false | 3       |
|     |      | 4   | 9   | false    | true  | 0       |
|     |      | 5   | 11  | false    | false | 1       |
|     |      | 5   | 12  | false    | true  | 0       |
|     |      | 6   | 13  | false    | false | 6       |
|     |      | 6   | 14  | false    | true  | 0       |
|     |      | 7   | 17  | false    | false | 1       |
|     |      | 7   | 18  | false    | true  | 0       |
|     |      | 8   | 20  | false    | false | 5       |
|     |      | 8   | 21  | false    | true  | 0       |
|     |      | 9   | 24  | false    | false | 4       |
|     |      | 9   | 25  | false    | true  | 0       |
|     |      | 10  | 28  | false    | false | 6       |
|     |      | 10  | 29  | false    | true  | 0       |
|     |      | 11  | 33  | false    | false | 1       |
|     |      | 11  | 34  | false    | true  | 0       |
|     |      | 12  | 37  | false    | false | 11      |
|     |      | 12  | 38  | false    | true  | 0       |
|     |      | 13  | 43  | false    | false | 1       |
|     |      | 13  | 44  | false    | true  | 0       |
|     |      | 14  | 48  | false    | false | 8       |
|     |      | 14  | 49  | false    | true  | 0       |
|     |      | 15  | 54  | false    | false | 6       |
|     |      | 15  | 55  | false    | true  | 0       |



```

13   v := for j in 0..3 list random(kk);
14   -- Ensure the point is not zero
15   while all(v, x -> x == 0) do (
16     v = for j in 0..3 list random(kk)
17   );
18   v
19 )
20 -- Pick nrPoints random points in P^3
21 myPoints = apply(nrPoints, randomPoints);
22 -- Create pairs of subsequent points, and also the pair consisting
    of the last and first point
23 subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i
    , myPoints#(i+1)};
24
25 -- Step 3: Generate matrices from point pairs
26 -- Function to convert two points to a matrix with first row {x_0
    .. x_3}
27 pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points
    #1};
28 -- Convert each pair of points to a matrix
29 pointsMatrix = apply(subsequentPairs, pointsToMatrix);
30
31 -- Step 4: Get the ideals of the lines
32 -- Given a matrix with first row {x_0 .. x_3} and next two rows
    points, return the ideal of the line through the points
33 lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
34 allLines = apply(pointsMatrix, lineFromPoints);
35
36 -- Step 5: Create the ideal of the union of all lines
37 -- Intersect the ideals of the lines to get ideal of the union of
    all lines
38 Iz = intersect allLines;
39
40 -- Step 6: Pick a degree d element in Iz, which defines a surface
    containing all the lines, and check properties
41 -- Pick a random degree d element in Iz
42 f = random(d, Iz);
43 -- Create the variety defined by f
44 Vf = variety ideal f;
45
46 -- Check if the surface is smooth and not the zero polynomial
47 codimension = codim singularLocus ideal f;
48 isZero = (f == 0)
49 isSingular = (codimension < 4)
50 isSmooth(Vf) and not isZero
51
52 F = sheaf(Iz);
53 H0 = HH^0(F(d))
54 H1 = HH^1(F(d))
55
56 -- primaryDecomposition(Iz)
57 -- factor(f)
58 maxNoPoints = numeric((binomial(d+3, 3) - 1)/d)

```

### A.3 Code for the extra interested

app:loop

To scan many  $(d, m)$  pairs automatically a while-loop script iterates degrees until a time budget is exhausted.

lst:m-chain

Listing A.3: Script for looping of different degrees of  $m$ -chain (number of lines =  $m - 1$ )

```

1 listOfResults = {};
2 listOfSurprises = {};
3
4 kk = ZZ/32749;
5 -- kk = QQ;
6 RingP3 = kk[x_0..x_3];
7
8 pointsMatrix = points -> matrix {{x_0 .. x_3}, points#0, points
  #1};
9 lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
10
11 time0 = currentTime();
12 d = 2;
13 while currentTime() - time0 < 60*60 do( -- wait for 1 hour
14   maxNoLines = (binomial(d+3, 3))/d;
15   print(numeric(maxNoLines));
16   maxNoPoints = ceiling((d+3)*(d+2)*(d+1)/(6*d));
17   for noPoints from maxNoPoints-1 to maxNoPoints do ( --
     maxNoPoints to maxNoPoints + 1 for m-chains, maxNoPoints-1
     to maxNoPoints for m-gons
18     myPoints = apply(noPoints, i -> (
19       v := for j in 0..3 list random(kk);
20       -- Ensure the point is not the zero vector
21       while all(v, x -> x == 0) do (
22         v = for j in 0..3 list random(kk)
23       );
24       v
25     ));
26     subsequentPairs = for i from 0 to (#myPoints - 2) list {
       myPoints#i, myPoints#(i+1)};
27     subsequentPairs = append(subsequentPairs, {myPoints#0,
       myPoints#(#myPoints-1)}); -- close the loop
28     pointPairs = apply(subsequentPairs, pointsMatrix);
29     allLines = apply(pointPairs, lineFromPoints);
30
31     Iz = intersect allLines;
32     f = random(d, Iz);
33
34     hf = hilbertFunction(d, RingP3/Iz);
35     -- Total number of degree-d monomials in 4 variables over
       kk is binomial(d+3,3)
36     totalForms = binomial(d+3,3);
37     numInIz = totalForms - hf; -- number of independent degree
       -d forms vanishing on Z
38
39     if f == 0 then (
40       if numInIz > 0 then (
41         tries = 0;
42         maxAttempts = 100; -- abort with an error if we

```

```

cannot find a nonzero element
43 while f == 0 do (
44   f = random(d, Iz);
45   tries = tries + 1;
46   if tries > maxAttempts then error ("Could not
      find nonzero element of degree " | toString
      d | " in Iz after " | toString maxAttempts
      | " attempts");
47   );
48 );
49 );
50
51 codimension = codim singularLocus ideal f;
52 isZero = (f == 0);
53 isSingular = (codimension < 4);
54 listOfResults = append(listOfResults, (d, noPoints, isZero
55   , isSingular, currentTime() - time0));
56 if isSingular then (
57   listOfSurprises = append(listOfSurprises, (d, noPoints
58     , isZero, isSingular, currentTime() - time0))
59 );
60 print("d = " | toString d | ", noPoints = " | toString
61   noPoints | ", isSingular = " | toString isSingular | ",
62   isZero = " | toString isZero | ", numInIz = " |
63   toString numInIz | ", time = " | toString (currentTime
64     () - time0) );
65 );
66 d = d + 1;
67 )
68
69 time1 = currentTime();
70 timeTaken = time1 - time0
71 -- listOfResults
72 listOfSurprises

```